

Non-Commutative Noetherian Ring

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Introduction

Here is the prologue of the notes by the lecturer:

This course is a first course in noncommutative rings. Familiarity with the theory of commutative rings will be assumed.

Rings arise in a variety of contexts throughout mathematics but one central context is as linear operators on a suitable space. These will rarely commute. As for commutative rings, noetherianity of a ring imposes a degree of control — indeed it is perhaps even more significant in the noncommutative setting in order to avoid pathological examples where very little can be said of a general nature

As this indicates, the theory of noncommutative rings is helpful if we need to understand the endomorphism ring of some objects, as often seen in the context of Galois representations. For this reason, this course is closely related to representation theory.

Here are the official lecture notes released by the lecturer: <https://www.dpmms.cam.ac.uk/~sjw47/2026Lectures.pdf>. This is my revision notes, though it didn't work as well as I imagined for the reason about to reveal. The main changes in the revision notes are the following:

1. The proofs of many theorems, including non-commutative version of Hilbert's Basis Theorem, are sketched in these notes, because I thought these proofs were too technical while conceptually not important for this course. I am totally wrong on this because two of them appeared as bookworks in the exam, and I was unable to recover the details.
2. The construction of the left ring of fractions w.r.t. a left Ore set is omitted. Chapter VI of *An Introduction to Noncommutative Noetherian Rings* by Goodearl/Warfield gives a more motivational and comprehensible presentation on this content.
3. Some proofs of exercises or lemmas are rewritten or simplified by me. I guess this is the whole point of rewriting these notes on one's own.

There is no short-cut to study noncommutative ring theory and many examples need to be familiarized with. But it is still a fun experience no matter what. (though I should have chosen Lie algebra for exam)

1 Basic Definitions

In this course rings are associative and unital, but are not necessarily commutative. We first need to get familiar with the context of non-commutative rings.

A **ring** $R = (R, +, \cdot)$ is an abelian group under $+$ with unit 0 such that $\cdot$ is a $\mathbb{Z}$ -bilinear associative binary operation with identity 1 .

Given a ring $(R, +, \cdot)$ we can define a ring $R^{op} = (R, +, *)$ with $a * b = b \cdot a$ for all $a, b \in R$. Of course if R is commutative then $R = R^{op}$. We use $R^\times$ to denote the units of R

The **center** of a ring R is the set $Z(R) = \{z \in R \mid zr = rz \ \forall r \in R\}$, which is always a subring of R

Example 1. $Z(Mat_n(k)) = kI_n$

A **(two-sided) ideal** I of a ring R is a subgroup of $(R, +)$ such that $axb \in I$ for all $x \in I$ and $a, b \in R$. We write $I \triangleleft R$. Given such an ideal I , R/I is a ring under operations induced from R . If I is a left- R -submodule with $ax \in I$ for $a \in R, x \in I$, it is called a **left ideal**, and we write $I \triangleleft_l R$

A ring homomorphism $\alpha : R \rightarrow S$ is as in the commutative case.

Remark 1.1. $\alpha(R^\times) \subset S^\times$, but $\alpha(Z(R)) \not\subset Z(S)$ in general because α is not surjective.

If k is a commutative ring, a **k -algebra** is a ring R equipped with a ring homomorphism $\iota_R : k \rightarrow Z(R)$.

For every ring R and set X we may form RX , the free R -module on X , as the set of finitely supported functions $f : X \rightarrow R$ under pointwise operations.

Remark 1.2. Note that we can write a function f in RX in the form of finite sum $\sum_{x \in X} a_x x$ by letting $a_x = f(x)$. This agrees with our normal definition of group algebra, which is more intuitive.

If M is a R -module for some k -algebra R , then the set $End_R(M) = Hom_R(M, M)$ is a k -algebra with group of units $Aut_R(M)$.

Example 2. If $M = \mathbb{Z}^n$, then $End(M) \cong M_n(\mathbb{Z})$

Remark 1.3. Every R -module structure on an abelian group M gives a ring homomorphism $\theta : R \rightarrow End(M)$ given by $\theta(r)(a) = r \cdot a$

Let G be a group and let k be a commutative ring. The **group algebra** kG is the k -algebra consisting of finitely supported functions $G \rightarrow k$ with pointwise addition and multiplication given by convolution:

$$f_1 f_2(g) = \sum_{h_1 h_2 = g} f_1(h_1) f_2(h_2)$$

Remark 1.4. Again, this definition of group algebra and its multiplication agrees with our normal definition.

Proposition 1.1. Suppose that R is a k -algebra. Restriction along the group homomorphism $G \rightarrow (kG)^\times, g \mapsto \delta_g$ defines a bijection

$$\Theta : Hom_k(kG, R) \rightarrow Hom_{gp}(G, R^\times)$$

Suppose that k is a field. Recall that a k -linear representation of G is a group homomorphism $\varphi : G \rightarrow Aut_k(V)$ where V is some vector space over k .

Corollary 1.1. There is a natural bijection between k -linear representations of G and left kG -modules.

Proof. This follows from remark 1.3 and proposition 1.1 immediately. $\square$

1.1 Universal Enveloping Algebra

A **Lie algebra** over k is a k -module $\mathfrak{g}$ equipped with a k -bilinear map $[\cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ (the Lie bracket) satisfying

1. $[xx] = 0$ for $x \in \mathfrak{g}$. Hence $[yz] = [zy]$ for $y, z \in \mathfrak{g}$ (anti-commutativity)
2. $[x[yz]] + [y[zx]] + [z[xy]] = 0$ for $x, y, z \in \mathfrak{g}$

Note this bracket is not associative.

Example 3. Any k -algebra R becomes a Lie algebra over k under the commutator bracket: $[xy] = xy - yx$

A **k -Lie algebra homomorphism** $\theta : \mathfrak{g}_1 \rightarrow \mathfrak{g}_2$ is a k -linear map such that $\theta([xy]) = [\theta(x), \theta(y)]$, i.e. the Lie structure is preserved.

A representation of the Lie algebra $\mathfrak{g}$ is a k -Lie algebra homomorphism $\mathfrak{g} \rightarrow End_k(V)$ for some k -module V . Here $End_k(V)$ is a Lie algebra with commutator bracket.

We now wish to find an analogue of the group algebra for representations of Lie algebras.

Definition 1.1. Suppose that V is a k -module. Define $T^0V = k$, $T^1V = V$, $T^nV = V \otimes_k T^{n-1}V$ for $n \geq 2$. The **tensor algebra** of V is $TV = \bigoplus_{n \geq 0} T^nV$ with multiplication given by $T^nV \times T^mV \rightarrow T^{n+m}V$, $v_1 \otimes \dots \otimes v_n, w_1 \otimes \dots \otimes w_m \mapsto v_1 \otimes \dots \otimes w_m$

If V is free k -module on $x_1, \dots, x_d$ we write $k\langle x_1, \dots, x_d \rangle = TV$ and call it the **associated algebra** on $x_1, \dots, x_d$. Note that it is not commutative for $d \geq 2$.

Proposition 1.2. Suppose that V is a k -module and R is a k -algebra. There is a natural bijection

$$\Theta : \text{Hom}_{k\text{-alg}}(TV, R) \rightarrow \text{Hom}_k(V, R)$$

by restricting to $T^1V = V$

Definition 1.2. Suppose $\mathfrak{g}$ is a k -Lie algebra. The **Universal Enveloping Algebra** $U(\mathfrak{g})$ of the Lie algebra $\mathfrak{g}$ is $U(\mathfrak{g}) = T\mathfrak{g}/I$ where I is the two sided ideal in $T\mathfrak{g}$ generated by $\{xy - yx - [xy] \mid x, y \in \mathfrak{g} = T^1\mathfrak{g}\}$. There is a natural map of Lie algebras $\iota_{\mathfrak{g}} : \mathfrak{g} \rightarrow U(\mathfrak{g})$ by $x \mapsto x + I$

Example 4. If $\mathfrak{g}$ is the abelian Lie algebra (i.e. Lie bracket is just zero) free on $x_1, \dots, x_d$, then $U(\mathfrak{g})$ is just the polynomial algebra $k[x_1, \dots, x_d]$. More generally when $V = \mathfrak{g}$ is abelian $U(V)$ is called the **symmetric algebra** on V and written SV . (abelianization)

Proposition 1.3. Suppose that $\mathfrak{g}$ is a k -Lie algebra and R is an (associative) k -algebra. There is a natural bijection

$$\Theta : \text{Hom}_{k\text{-alg}}(U(\mathfrak{g}), R) \rightarrow \text{Hom}_{k\text{-Lie}}(\mathfrak{g}, R)$$

where R is equipped with the commutator bracket by restricting along $\iota_{\mathfrak{g}}$

Corollary 1.2. There is a natural bijection between representations of the Lie algebra $\mathfrak{g}$ and left $U(\mathfrak{g})$ -modules.

1.2 Weyl Algebra

Recall that a **k -derivation** of a k -algebra R is a k -linear map $\delta : R \rightarrow R$ such that $\delta(1) = 0$ and $\delta(rr') = r\delta(r') + \delta(r)r'$. We write $\text{Der}_k(R)$ for the set of k -derivations of R .

Remark 1.5. $\text{Der}_k(R)$ is a k -Lie algebra with commutator bracket.

Definition 1.3. Let k be a field and let A be a commutative k -algebra. Let $\Delta(A)$ be the k -subalgebra of $\text{End}_k(A)$ generated by the k -linear operators $\hat{a} : b \mapsto ab$ and $\text{Der}_k(A)$. We call $\Delta(A)$ the **derivation ring** A .

Definition 1.4. Let k be a field. The n -th Weyl algebra $A_n(k)$ over k is

$$A_n(k) = k\langle x_1, \dots, x_n, y_1, \dots, y_n \rangle / I$$

where I is the ideal generated by $x_i x_j - x_j x_i$, $y_i y_j - y_j y_i$, $y_i x_i - x_i y_i - 1$, $y_i x_j - x_j y_i$ for $i \neq j$.

Lemma 1.1. Let $\hat{x}_i : f \mapsto x_i \cdot f$. There is a surjective k -algebra homomorphism from $A_n(k)$ to $\Delta(k[x_1, \dots, x_n])$ by sending x_i to $\hat{x}_i$ and y_i to $\partial/\partial x_i$

We see that $\hat{x}_i$ and $\partial/\partial x_i$ fit into these relations, so this is by the universal property of quotient.

Remark 1.6. For most $A_n(k)$ -modules, we will see that x_i acts by $\hat{x}_i$ and y_i acts by $\partial/\partial x_i$. So we should treat x_i, y_i as $\hat{x}_i$ and $\partial/\partial x_i$ since this motivates the relations in the definition.

1.3 Exercise

Here is a little trick we will meet frequently

Lemma 1.2. *Let R be a ring and $x \in R$. Define $ad_x(r) = xr - rx$. we have*

$$x^n r = \sum_{i=0}^n \binom{n}{i} ad_x^i(r) x^{n-i}$$

for all $r \in R$

Proof. Consider two $R \rightarrow R$ maps: $l_x : r \mapsto xr$, $r_x : r \mapsto rx$. Note that $ad_x = l_x - r_x$, and these three operators commute.

$$\text{Hence } x^n r = l_x^n(r) = (ad_x + r_x)^n r = \sum_{i=0}^n \binom{n}{i} ad_x^i \circ r_x^{n-i}(r) = \sum_{i=0}^n \binom{n}{i} ad_x^i(r) x^{n-i} \quad \square$$

We will use this lemma to show certain multiplicative closed set S is left-Ore, by showing that ad_x is nilpotent for $x \in S$. Then for n large $RHS = bx$ for some $b \in R$

2 Noetherianity

Noetherian condition is a key finiteness property of modules that is sufficiently restrictive to allow strong structure theorems to be proved, and yet sufficiently general to include large classes of interesting noncommutative rings.

The (left) R -module M is said to be **cyclic** if it can be generated by a single element $M = Rx$ for some $x \in M$. It is **finitely generated** if it can be written as a finite sum of cyclic submodules.

Lemma 2.1. *Let M be an R -module. TFAE:*

1. *Every submodule of M is finitely generated*
2. *Ascending chain condition (acc): There does not exist an infinite strictly ascending chain of submodules of M*
3. *Maximum condition: Every non-empty subset of submodules of M contains at least one maximal element.*

This is exactly the same as for the commutative case.

An R -module satisfying lemma 2.1 is **noetherian**. The ring R is left noetherian if it is noetherian as a left R -module. An R -module satisfying the descending chain condition, or equivalently, the minimum condition, is said to be **artinian**. The ring R is left artinian if it is artinian as a left R -module.

Here is a key result for proving that certain rings are left noetherian: it is a non-commutative version of *Hilbert's Basis Theorem*.

Theorem 2.1. *Let S be a ring, R a left noetherian subring and suppose that for some $x \in S$ we have*

1. $R + xR = R + Rx$, and
2. $S = \langle R, x \rangle$

Then S is also left noetherian.

The proof is very technical. First we can prove by induction that $R + xR + \dots + x^n R = R + Rx + \dots + Rx^n$ for all n by (1) and denote it by S_n . Then we observe that $S = \cup_{n \geq 0} S_n$ since it is a subring of S containing R and x and $S_n S_m = S_{n+m}$. Then for $I \triangleleft_l S$, we try to descend it to a series of left ideals I_n of R and fiddle with their generators to get generators of I . This is much harder than it sounds like. Details can be found in lecturer's notes.

Definition 2.1. A group G is said to be **polycyclic** if there is a chain of subgroups such that each quotient is cyclic.

Example 5. $\{I + N \in M_n(\mathbb{Z})\}$ where N is strictly upper triangular is always polycyclic.

Theorem 2.2. (Auslander-Swan). Let G be a polycyclic group. Then

1. G is isomorphic to a subgroup of $GL_n(\mathbb{Z})$ for some n .
2. G has a subgroup H of finite index isomorphic to a subgroup of $Tr_n(\mathcal{O})$, the group of upper-triangular matrices in $GL_n(\mathcal{O})$, where $\mathcal{O}$ is the ring of integers of an algebraic number field.

Proposition 2.1. Let k be a commutative noetherian ring and let G be a polycyclic group. Then kG is noetherian.

Although these theorems look nice, they are not needed for the rest of this course.

Remark 2.1. $U(\mathfrak{g})$ is noetherian whenever $\mathfrak{g}$ is a finitely generated k -module and k is noetherian, which is to be seen.

Weyl algebras $A_n(k)$ for k a field are also noetherian. To see this, we introduce the **skew polynomial ring** $S = R[x; \delta]$ over R , where S is a free left- R -module with basis $x^i, i \geq 0$ and δ is a derivation on R satisfying $xr = rx + \delta(r)$ for all $r \in R$. Then S is left noetherian if R is left noetherian by Hilbert's basis theorem.

If we take $R = k[y]$ and $\delta = \partial/\partial y$, then $S \cong A_1(k)$ is noetherian since $k[y]$ is noetherian. In general, notice that $A_n(k) = k[x_1, \dots, x_n][y_1, \partial/\partial x_1][y_2, \partial/\partial x_2] \dots [y_n, \partial/\partial x_n]$ by letting $\partial/\partial x_j(y_i) = 0$ for $i < j$. Apply the observation above n times and we see that $A_n(k)$ is noetherian.

Non-commutative finitely generated k -algebras are not always noetherian!

2.1 Exercise

First we give several examples of non-commutative rings that are not (left or right) noetherian.

Lemma 2.2. The free algebra $k\langle x, y \rangle$ is neither left nor right noetherian. Deduce that $k\langle x_1, \dots, x_n \rangle$ is neither left nor right noetherian when $n \geq 2$.

Proof. For $k\langle x, y \rangle$, let $I_n = \sum_{i=0}^n k\langle x, y \rangle yx^i$. Then $yx^{n+1} \notin I_n$ and $yx^{n+1} \in I_{n+1}$. So $\{I_n\}$ is a strictly ascending chain of left ideals. Similarly, let $J_n = \sum_{i=0}^n x^n y \cdot k\langle x, y \rangle$. Hence $k\langle x, y \rangle$ is neither left nor right noetherian.

For $n \geq 2$, consider $\phi : k\langle x_1, \dots, x_n \rangle \rightarrow k\langle x, y \rangle$ given by $x_1 \mapsto x, x_i \mapsto y$ for $i \geq 2$. Then $k\langle x, y \rangle$ is a finitely generated but non-noetherian $k\langle x_1, \dots, x_n \rangle$ -module on both sides. □

Next are examples of rings that are right, but not left, noetherian, and vice versa.

Example 6. $\begin{pmatrix} \mathbb{Z} & \mathbb{Q} \\ 0 & \mathbb{Q} \end{pmatrix}$ is right, but not left, noetherian. Indeed, $\begin{pmatrix} 0 & \frac{1}{n}\mathbb{Z} \\ 0 & 0 \end{pmatrix}$ is a collection of left ideals with no maximal element. On the other hand, we have $\begin{pmatrix} 0 & \mathbb{Q} \\ 0 & \mathbb{Q} \end{pmatrix} \subset \begin{pmatrix} \mathbb{Z} & \mathbb{Q} \\ 0 & \mathbb{Q} \end{pmatrix}$ as a right submodule. It is noetherian since $\mathbb{Q}^2$ is noetherian as $\mathbb{Q}$ -module. Note that $\begin{pmatrix} \mathbb{Z} & \mathbb{Q} \\ 0 & \mathbb{Q} \end{pmatrix} / \begin{pmatrix} 0 & \mathbb{Q} \\ 0 & \mathbb{Q} \end{pmatrix} \cong \mathbb{Z}$ as a right $\begin{pmatrix} \mathbb{Z} & \mathbb{Q} \\ 0 & \mathbb{Q} \end{pmatrix}$ -module via $n \cdot \begin{pmatrix} m & a \\ 0 & b \end{pmatrix} = nm$. Hence the quotient is also noetherian. Similarly, we can show $\begin{pmatrix} \mathbb{Q} & \mathbb{Z} \\ \mathbb{Q} & 0 \end{pmatrix}$ is left but not right noetherian

Theorem 2.3. Let R be a ring and let M be a noetherian left R -module. Suppose $\varphi : M \rightarrow M$ is an R -module homomorphism. Show that if φ is surjective, then it is also injective. Deduce that if R is a left noetherian ring with $ab = 1$ for some $a, b \in R$ then $ba = 1$.

Proof. We have an ascending chain $\ker\varphi \subset \ker\varphi^2 \subset \dots$ of left R -submodules of M . Since M is left noetherian, $\ker\varphi^n = \ker\varphi^{n+1}$ for some n . Let $x \in \ker\varphi$. φ is surjective implies that φ^n is surjective, so there is $y \in M$ such that $\varphi^n(y) = x$, and so $\varphi^{n+1}(y) = \varphi(x) = 0 \implies y \in \ker\varphi^{n+1} = \ker\varphi^n$. So $x = \varphi^n(y) = 0$.

Now for R a left noetherian ring and $a, b \in R$ with $ab = 1$, $\phi : R \rightarrow R; r \mapsto rb$ is a left R -module homomorphism with $\phi(a) = ab = 1$. So ϕ is surjective and hence injective by preceding result. Now note that $\phi(ba) = bab = b(ab) = b = \phi(1)$, so $ba = 1$ $\square$

Theorem 2.4. Suppose k is a field of characteristic 0. Recall that $\Delta(A)$ is the derivation ring of A .

1. Show that $k[x]$ is a simple $\Delta(k[x])$ -module under the natural action. Show that $k[x, x^{-1}]$ is also naturally a $\Delta(k[x])$ -module. Determine its submodules and quotient modules.
2. Consider the k -Lie algebra sl_2 of 2×2 -matrices with trace 0 and its basis $e = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, f = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. Then for each $\lambda \in k$ there is a unique k -algebra homomorphism $\theta_\lambda : U(sl_2(k)) \rightarrow \Delta(k[x])$ given by $e \mapsto -d/dx, h \mapsto -2\hat{x}d/dx + \lambda, f \mapsto \hat{x}^2d/dx - \lambda\hat{x}$
3. When is $k[x]$ a simple $U(sl_2(k))$ -module under the action induced by θ_λ ? In the cases when it is not simple what are its submodules and quotient modules? What about for $k[x, x^{-1}]$?

Proof. (1) Take $0 \neq f \in k[x]$ with degree n . Then $(d/dx)^n f = \lambda \neq 0 \in k$. Then λ generates $k[x]$ as $\Delta(k[x])$ -module (recall that $\theta(a) : b \mapsto ab \in \Delta(k[x])$ for all $a \in k[x]$). Similarly we can show that $k[x, \frac{1}{x}]/k[x]$ is a simple $\Delta(k[x])$ -module.

(2) By proposition 1.3 (universal property of universal enveloping algebra), it suffices to show that this gives a k -Lie algebra homomorphism $sl_2 \rightarrow \Delta(k[x])$. For example, $\theta_\lambda([e, h]) = \theta_\lambda(-2e) = 2d/dx$, while $[\theta_\lambda(e), \theta_\lambda(h)] = [-d/dx, -2\hat{x}d/dx + \lambda] = 2[d/dx, \hat{x}]d/dx = 2d/dx$

(3) For $0 \neq p \in k[x]$ with degree n , then $e^n p \in k$ and so 1 is in the module it generates. Also note that $fx^i = ix^{i+1} - \lambda x^{i+1}$. Thus if $\lambda \notin \mathbb{N}$, then fx^i is never 0 and we can produce x^i for all i in the $U(sl_2(k))$ -submodule generated by p . So $k[x]$ is simple.

If $\lambda \in \mathbb{N}$, then $fx^\lambda = 0$. Then $V = (1, x, \dots, x^\lambda)$ is a $U(sl_2(k))$ -submodule and $k[x]/V$ is simple. $\square$

3 Simple modules, Primitive Ideals and Jacobian Radical

An R -module M is simple or irreducible if $M \neq 0$ and the only submodules of M are 0 and M .

Suppose M is simple. Take $0 \neq x \in M$. Then $M = Rx$, and so $M \cong R/\text{ann}_R(x)$, where $\text{ann}_R(x)$ is the **left-annihilator** of x . Then it is easy to see that $M = Rx$ is simple if and only if $\text{ann}_R(x)$ is a maximal left ideal.

Lemma 3.1. (*Zorn's Lemma*). *Let $(S, \leq)$ be a poset. If every totally ordered subset (chain) of S has an upper bound then, for every $s \in S$, S has a maximal element x with $s \leq x$.*

This is equivalent to the Axiom of Choice, which we will always assume holds.

As in the commutative case we can use Zorn's Lemma to show that every left ideal I of a ring R is contained in a maximal left ideal.

Lemma 3.2. *Suppose L is a proper left ideal of R . Then L is contained in a maximal left ideal I of R .*

Corollary 3.1. *Every nonzero finitely generated R -module M has a simple quotient (as an R -module).*

We can take a smallest generating set $x_1, \dots, x_n$ of M and let $N = \sum_{i=1}^{n-1} Rx_i$. Then $M/N \cong R/\text{ann}_R(x_n + N)$ is generated by $x_n + N$ and cyclic. By lemma 3.2 we can take a maximal ideal I containing $\text{ann}_R(x_n + N)$, which corresponds to a submodule M' with $M/M' \cong R/I$ and hence this quotient is simple.

Remark 3.1. Finitely generated condition is necessary. For example, $\mathbb{Q}$ has no simple quotient as a $\mathbb{Z}$ -module.

Note that the (left)-annihilator $\text{Ann}_R(M) = \cap_{m \in M} \text{ann}_R(m)$ is a two-sided ideal of R . An R -module M is **faithful** if $\text{Ann}_R(M) = 0$, or equivalently, the natural map $R \rightarrow \text{End}_R(M)$ is injective.

Definition 3.1. *Let I be a two-sided ideal of R . Then I is left **primitive** if I is the annihilator of a simple left R -module M .*

*The ring R itself is called left **primitive** if its zero ideal is left primitive, or equivalently, if R has at least one **faithful simple left module**.*

Lemma 3.3. *Let $M = Rx$ be a cyclic left R -module. Then $\text{Ann}_R(M)$ is the largest two-sided ideal contained in $L = \text{ann}_R(x)$.*

Let $I \subset L$ be a two-sided ideal. Then $IM = IRx \subset Ix \subset Lx = 0$ and so $I \subset \text{Ann}_R(M)$

Corollary 3.2. *Every maximal ideal $I \triangleleft R$ is left and right primitive.*

Indeed, R/I is a two-sided simple R -module and $\text{Ann}_R(R/I) = I$

Definition 3.2. *The Jacobson radical $J(R)$ of R is defined to be the intersection of all left primitive ideals of R . Note that $J(R)$ is the set of elements of R which annihilate every simple left R -module, i.e. $J(R)$ kills every simple left R -module*

Lemma 3.4. *$J(R)$ equals to the intersection K of all left maximal ideal in R .*

Corollary 3.2 gives $J(R) \subset K$. Now suppose $P = \text{Ann}_R(M)$ is a left primitive ideal for some simple R -module M . Note that for each $0 \neq x \in M$ $\text{ann}_R(x)$ is a left maximal ideal in R since $M = Rx \cong R/\text{ann}_R(x)$ is simple. So $P = \cap_{0 \neq x \in M} \text{ann}_R(x)$ is an intersection of left maximal ideal and hence $K \subset P$. So $K \subset J(R)$

We still have Nakayama's lemma for Jacobson radical:

Lemma 3.5. (*Nakayama's lemma*) Let M be a finitely generated nonzero left R -module. Then $J(R)M$ is strictly contained in M .

Corollary 3.1 gives a simple quotient M/N of M . Then $J(R)(M/N) = 0$ and hence $J(R)M \subset N$. Since N is strictly contained in M , $J(R)M$ is strictly contained in M .

As in the commutative case, we use Nakayama's lemma to show finitely generated $M = 0$ if we can get $J(R)M = M$.

Corollary 3.3. Let M be a finitely generated left R -module. If N is a submodule of M such that $M = N + J(R)M$ then $M = N$.

Proof. Apply Nakayama's lemma to M/N □

Here is an important characterization of $J(R)$

Proposition 3.1. $J(R) = \{x \in R \mid 1 - axb \in R^\times \forall a, b \in R\} =: K$

Proof. Since $J(R)$ is a two-sided ideal, to show that $J(R) \subset K$ it suffices to show that for $x \in J(R)$, $1 - x \in R^\times$. Note that $R = R(1 - x) + Rx$. Since $Rx \subset J(R)$, corollary 3.3 gives $R = R(1 - x)$. So there exists $y \in R$ such that $y(1 - x) = 1$, i.e. we find the left inverse.

Note that $1 - y = -xy \in J(R)$. So we can repeat this for $1 - (1 - y) = y$ and get $z \in R$ such that $zy = 1$. Hence $z = zy(1 - x) = 1 - x$. So y is a two sided inverse and $1 - x \in R^\times$.

Now let $x \in K$. Suppose there exists a maximal left ideal I with $x \notin I$. Then, since I is maximal, $I + Rx = R$, so $1 - ax \in I$ for some $a \in R$. But this implies that $1 - ax$ is not a unit since I is a proper ideal, contradiction. So x is in the intersection of every maximal left ideal which is $J(R)$ by lemma 3.4. □

This result shows that $J(R)$ is the largest two-sided ideal I of R such that $1 - I$ consists entirely of units of R .

Corollary 3.4. As sets $J(R) = J(R^{op})$, it follows that the intersection of all maximal left ideals of R is equal to the intersection of all maximal right ideals.

Let V be a left R -module and let $D = \text{End}_R(V)^{op}$. We can view V as a right D -module via $V \times D \rightarrow V; (v, \phi) \mapsto v\phi = \phi(v)$

Lemma 3.6. (*Schur's lemma*) Let V be a simple left R -module. Then $D = \text{End}_R(V)$ is a division ring.

So whenever V is a simple left R -module, V becomes a right *vector space* over the division ring $D = \text{End}_R(V)$.

Lemma 3.7. Let V be a simple left R -module, let $D = \text{End}_R(V)$, let X be a finite subset of V and let $I = \text{ann}_R(X) = \cap_{x \in X} \text{ann}_R(x)$.

Suppose $I \cdot y = 0$ for some $y \in V$. Then $y \in X \cdot D$, the D -span of X in V .

The lemma says that the elements in V I annihilates are exactly the D -span of X

Proof. We proceed by induction on $n = |X|$. $n = 0$ is trivial. Assume now for $n \geq 1$ and let $J = \text{ann}_R(X \setminus \{x\})$ for some $x \in X$ and hence $I = J \cap \text{ann}_R(x)$. If $J \subset \text{ann}_R(x)$ then $I = J$ and $J \cdot y = 0$, so we can apply inductive hypothesis. So we can assume that J is not contained in $\text{ann}_R(x)$. But then the R -submodule $J \cdot x$ of V is non-zero, so $J \cdot x = V$ by the simplicity of V .

Hence we can define $d : V \rightarrow V$ by $(r \cdot x)d = r \cdot y$ for $r \in J$. This is well-defined because if $r \cdot x = 0$, then $r \in \text{ann}_R(x) \cap J = I$ and so $r \cdot y = 0$ by assumption. It is clearly R -linear, so $d \in D$. So $J \cdot (y - x \cdot d) = 0$ and by induction, $y - x \cdot d \in X \setminus \{x\} \cdot D$, so $y \in X \cdot D$ □

Theorem 3.1. (*Jacobson's Density*). Let V be a simple left R -module, and let $X \subset V$ be a finite D -linearly independent subset of V where $D = \text{End}_R(V)$. Then for any $\alpha \in \text{End}_R(V)$, there exists $r \in R$ such that $\alpha(x) = rx$ for every $x \in X$, i.e. α acts by multiplication by scalar over the D -span of X . Moreover, if V is of finite dimension over D , we can take X to be its basis and get that each $\alpha \in \text{End}_R(V)$ acts by multiplication by a scalar $r \in R$.

Proof. Write $X = \{x_i, i = 1, \dots, n\}$ and $X_i = X \setminus \{x_i\}$. Since $x_i \notin X_i \cdot D$, lemma 3.7 gives $\text{ann}(X_i) \cdot x_i \neq 0$. So there exists some $r_i \in \text{ann}(X_i)$ such that $r_i \cdot x_i \neq 0$. Since V is simple, $R \cdot r_i \cdot x_i = V$, so we can find $s_i \in R$ such that $s_i \cdot r_i \cdot x_i = \alpha(x_i)$.

Now note that $\sum_j s_j r_j \cdot x_i = s_i r_i \cdot x_i = \alpha(x_i)$ since $x_i \in X_j$ for $i \neq j$ and so is killed by r_j . So we can take $\sum_j s_j r_j$ to be r . $\square$

Corollary 3.5. Let R be a left artinian ring, let V be a simple left R -module and let $D = \text{End}_R(V)^{\text{op}}$ (we have forgotten to add "op" in D across this section, but I guess it is fine.). Then V is finite dimensional as a right D -vector space.

Proof. Consider the set of left ideals $\{\text{ann}_R(X) \mid X \subset V, |X| < \infty\}$. Since R is left artinian, it has a minimal element $I = \text{ann}_R(X)$. WTS $V = X \cdot D$. Let $y \in V$ and consider $\text{ann}_R(X \cup \{y\}) \subset \text{ann}_R(X)$. Then minimality gives $I = \text{ann}_R(X \cup \{y\})$. So $I \cdot y = 0$. By lemma 3.7, $y \in X \cdot D$. So $V = X \cdot D$ $\square$

Now we want to prove the celebrated **Wedderburn's theorem**.

Lemma 3.8. Let S be a ring, let N be a right S -module and let $n \geq 1$ be an integer. Then the ring of endomorphisms of N^n as a right S -module is isomorphic to $\text{Mat}_n(\text{End}_S(N))$, i.e. $\text{End}_S(N^n) \cong M_n(\text{End}_S(N))$

Theorem 3.2. (*Artin-Wedderburn*). Let R be a left primitive, left artinian ring. Then $R \cong M_n(D)$ for some division ring D and integer $n \geq 1$.

Proof. Let V be a faithful simple R -module, and let $D = \text{End}_R(V)$. Then D is a division ring. Since R is left artinian, $V \cong D^n$ for some n by corollary 3.5. Also $D \cong \text{End}_D(D)$ as right D -module (this is just $A \cong \text{Hom}_A(A, A)$). So $\text{End}_D(V) = \text{End}_D(D^n) \cong M_n(\text{End}_D(D)) \cong M_n(D)$ by lemma 3.8.

Now we have a natural ring homomorphism $\phi : R \rightarrow \text{End}_D(V)$ given by $\phi(r)(v) = r \cdot v$. It is injective because V is faithful, and it is surjective by Jacobson's theorem. We conclude that $R \cong M_n(D)$. $\square$

Lastly we recall the Chinese remainder theorem

Theorem 3.3. (*Chinese Remainder Theorem*) Let R be a ring, and let $P_1, \dots, P_n$ be coprime two-sided ideals in R , i.e. $P_i + P_j = R$ for $i \neq j$. Then

$$R/(P_1 \cap \dots \cap P_n) = R/P_1 \times \dots \times R/P_n$$

3.1 Exercise

Proposition 3.2. Let R be a left artinian ring and let $x \in R$ be such that $sx = 0$ implies $s = 0$. Prove that x is a unit in R . Deduce that a left artinian ring without zero-divisors must be a division ring.

Proof. By considering the descending chain $(x) \supset (x^2) \supset \dots$ we get a left inverse y of x . Also $sy = 0 \implies s = syx = 0$. Now considering the descending chain $(y) \supset (y^2) \supset \dots$ we get a left inverse z of y . So $z = zyx = x$. So x is a unit. $\square$

Proposition 3.3. *Use the Jacobson Density Theorem to show that if R is left primitive but not left artinian then there is a division ring D such that for every natural number n there is a subring R_n of R with a surjective ring homomorphism $R_n \rightarrow M_n(D)$.*

Proof. Let M be the faithful simple R -module and $D = \text{End}_R(M)^{\text{op}}$. If M is finite dimensional over D , then we are back in the settings of Wedderburn's so suppose not. For each n pick $m_1, \dots, m_n \in M$ D -linearly independent elements. Let $R_n = \{r \in R \mid rm_i \in \sum_{j=1}^n m_j D \ \forall i\}$ which is a subring of R . Then the natural map $R_n \rightarrow \text{End}_D(\sum_{j=1}^n m_j D) \cong M_n(D); r \mapsto r \cdot (-)$ is surjective by Jacobson's Density theorem. $\square$

Proposition 3.4. *Let R be a finite dimensional k -algebra, where k is an algebraically closed field and let M be a simple R -module. Show that $\text{End}_R(M) \cong k$.*

Proof. By Schur's lemma $\text{End}_R(M)$ is a division ring. Also $M \cong R/\text{ann}_R(m)$ for any $0 \neq m \in M$. This implies that M is also finite dimensional over k since R is f.d. over k . Now for $0 \neq \phi \in \text{End}_R(M)$, since k is algebraically closed, ϕ admits an eigenvalue $\lambda \in k$ as a k -linear map, and so $\phi - \lambda \text{id}_M$ has non-trivial kernel in M , which must be the whole M since M is simple. So $\phi = \lambda \text{id}_M$ and $\text{End}_R(M) = k$ $\square$

Proposition 3.5. *Let k be a field of characteristic 0 and consider the k -Lie algebra $\mathfrak{g}$ with basis x, y and such that $[yx] = x$.*

1. *Show that $U(\mathfrak{g})$ is both left and right noetherian*
2. *Show that $U(\mathfrak{g})$ is not a simple ring.*
3. *Show that $U(\mathfrak{g})/U(\mathfrak{g})(x-1)$ is a simple faithful $U(\mathfrak{g})$ -module. Deduce that $U(\mathfrak{g})$ is a primitive ring.*

Proof. (1) In $U(\mathfrak{g})$ we have $yx - xy = [yx] = x$. So $U(\mathfrak{g})$ is the skew polynomial ring $k[x][y; \delta]$ with $\delta = x d/dx$. So $U(\mathfrak{g})$ is left noetherian by Hilbert's basis theorem like Weyl algebras.

(2) Note that $(x) \subsetneq U(\mathfrak{g})$ is a two-sided ideal so $U(\mathfrak{g})$ not simple. (Not true for $A_1(k)$, because $1 = yx - xy \in (x)$)

(3) Note that $U(\mathfrak{g})$ has k -basis $y^j x^i, i, j \geq 0$. Let $v = 1 + U(\mathfrak{g})/(x-1)U(\mathfrak{g})$. Then $U(\mathfrak{g})/U(\mathfrak{g})(x-1) = \bigoplus_{i \geq 0} k y^i v$. Moreover, $xy - yx = -x \implies xy = (y-1)x \implies xy^m = (y-1)^m x$. So in $U(\mathfrak{g})/U(\mathfrak{g})(x-1)$, $xy^m v = (y-1)^m x v = (y-1)^m v \implies (x-1)y^m v \in m y^{m-1} v + \text{lower degree terms in } y$.

So if we pick $0 \neq f \in k[y]$ with smallest degree such that $f v \in M \subset U(\mathfrak{g})v$. If f is not constant, we can lower its degree since $(x-1)f v \in M$ and $(x-1)f$ has smaller degree in y in $U(\mathfrak{g})/U(\mathfrak{g})(x-1)$ by the above observation. So f is constant and $v \in M$ and so $M = U(\mathfrak{g})v = U(\mathfrak{g})/U(\mathfrak{g})(x-1)$. This shows that $U(\mathfrak{g})/U(\mathfrak{g})(x-1)$ is simple.

Let $\alpha = \sum y^i P_i(x) \in \text{Ann}_{U(\mathfrak{g})}(U(\mathfrak{g})/U(\mathfrak{g})(x-1))$. If $\alpha \neq 0$, i.e. some $P_i(x) \neq 0$, consider the largest power $(x-1)^m$ dividing every $P_i(x)$. Note that $P_i(x)y^m v = 0$ if $(x-1)^{m+1} \mid P_i(x)$ and $P_i(x)y^m v = \text{some nonzero constant}$ if $(x-1)^m \mid P_i(x)$ but $(x-1)^{m+1} \nmid P_i(x)$. From this we see that $\alpha y^m v \neq 0$ $\square$

4 Semiprimitive Rings and Semisimple modules

A ring R is **semi-primitive** if $J(R) = 0$ (recall that for a primitive ring, the zero ideal is primitive. So primitive rings are semi-primitive).

We call a non-zero ring with no non-zero proper two-sided ideals **simple**.

Example 7. If D is a division ring, then $M_n(D)$ is simple. Let E_{ab} be the matrix with (i, j) -th entry 1 and 0 elsewhere. Suppose non-zero $A \in M_n(D)$ with $a_{ij} \neq 0$. Then $a_{ij}^{-1}(E_{ki} A E_{jl}) = E_{kl} \in \langle A \rangle$ for any k, l . This implies $\langle A \rangle = M_n(D)$.

Now we classify semi-primitive rings using Wedderburn's.

Corollary 4.1. *Let R be a semi-primitive left artinian ring. Then there exist division rings $D_1, \dots, D_n$ and integers $d_1, \dots, d_n \geq 1$ such that*

$$R = M_{d_1}(D_1) \times \dots \times M_{d_n}(D_n)$$

Proof. Let S be the set of finite intersections of left primitive ideals of R ; it is non-empty since maximal ideals are primitive. Since R is left artinian, let $I = P_1 \cap \dots \cap P_n$ be a minimal element. For another primitive ideal Q , minimality gives $I = I \cap Q$ so $I \subset Q$. This implies that $I \subset J(R) = 0$. Now R/P_i is left primitive and left artinian and hence $R/P_i \cong M_{d_i}(D_i)$ for some division ring D_i by Wedderburn's. But example 6 shows that $M_{d_i}(D_i)$ is simple, so P_i is maximal. Then Chinese Remainder theorem gives $R = R/I = R/P_1 \times \dots \times R/P_n = M_{d_1}(D_1) \times \dots \times M_{d_n}(D_n)$ $\square$

Remark 4.1. In the proof we showed that for semi-primitive left artinian ring, an ideal is maximal if and only if it is left primitive.

Conversely, $R = M_{d_1}(D_1) \times \dots \times M_{d_n}(D_n)$ is semiprimitive left artinian, which can be seen from example 7 as well.

Definition 4.1. A (left) R -module M is **semisimple** if M is a direct sum of simple R -modules: $M = \oplus_{i \in I} M_i$

Note that a ring R is semiprimitive if and if R has a faithful semisimple module (direct sum all simple modules).

Lemma 4.1. *Any subquotient N of a semisimple module M is semisimple.*

It suffices to show that any submodule and quotient of M is semisimple. Let N be a submodule and $M = \oplus_{i \in I} M_i$. With Zorn's lemma, we can show that there exists a maximal sub-index $J \subset I$ such that $N \oplus (\oplus_{j \in J} M_j) = M$. Then this is clear.

Remark 4.2. Every submodule of a semisimple module M is a direct summand of M . Indeed this gives an equivalent definition of semisimple module, i.e. M is semisimple if every submodule is a direct summand. This implies that M has no proper essential submodule.

Lemma 4.2. *Let M be a semisimple R -module. TFAE*

1. M is noetherian
2. M is artinian
3. M is a **finite** direct sum of simple R -modules.

Since simple modules are noetherian and artinian, (3) $\implies$ (1), (2) is clear. Conversely, suppose M has a submodule $N = \oplus_{n \geq 1} M_n$ with M_i simple, we can construct a strictly ascending chain $\{\oplus_{n=0}^i M_n, i = 1, 2, \dots\}$ and a strictly descending chain $\{\oplus_{n \geq i} M_n, i = 1, 2, \dots\}$.

Remark 4.3. R is semiprimitive and left artinian if and only if R is a semi-simple R -module if and only if all left (right) R -modules are semisimple.

We conclude this section by proving some general results about left artinian rings as in the commutative case.

Proposition 4.1. *The Jacobson radical of a left artinian ring R is nilpotent.*

First we have a descending chain $J \supset J^2 \supset \dots$ and so $J^n = J^{n+1}$ for n sufficiently large. Consider I the ideals of elements killed by J^n . If $I \subsetneq R$, then R/I has a minimal non-zero left R -submodule K/I since R/I left artinian. Then K/I is a simple R -module and so $J \cdot K/I = 0$. Hence $JK \subset I$ and $J^n K = J^{n+1} K \subset J^n I = 0$ which implies that $K \subset I$, a contradiction. So $I = R$ and $J^n = 0$

Remark 4.4. This shows that for a left artinian ring $J(R) = N(R)$, the maximal nilpotent ideal in R .

Theorem 4.1. (*Hopkins–Levitski, 1939*). *Let R be a left artinian ring. Then R is also left noetherian.*

Let $J = J(R)$. Then J^i/J^{i+1} is a left artinian R -module. Since R/J is semiprimitive and left artinian, J^i/J^{i+1} is semisimple as R/J -module and hence as R -module by remark 4.2. Then lemma 4.3 shows that J^i/J^{i+1} is a finite direct sum of simple R -module and left noetherian. Since J is nilpotent, R is a finite extension of left noetherian R -modules, and is therefore itself left noetherian.

4.1 Exercise

Proposition 4.2. *TFAE:*

1. *All left R -modules are semisimple;*
2. *R is a semisimple left- R -module*
3. *R is semiprimitive and left artinian*

Proof. (1) $\implies$ (2) is clear.

(2) $\implies$ (1): If M is any R -module, consider the free R -module on M $RM \cong \bigoplus_{m \in M} R \rightarrow M; 1_m \mapsto m$ which is surjective. Since R is semisimple, RM is semisimple and hence M is semisimple by lemma 4.1 as it is a quotient of semisimple module

(2) $\implies$ (3): Suppose $R = \bigoplus_{i \in I} M_i$ with M_i simple. WLOG let $1 = \sum_{i=1}^m d_i$ with $0 \neq d_i \in M_i$. Then $M_i = Rd_i$ and so $R = \bigoplus_{i=1}^m M_i$. $J(R) = 0$ since $J(R)R = 0$. Now use lemma 4.2

(3) $\implies$ (2) is by corollary 4.1 □

Proposition 4.3. *Let k be a field, and let $T := \{A \in M_n(k) : a_{ij} = 0 \text{ whenever } i > j\}$ be the ring of upper-triangular matrices with entries in k . Find the Jacobson radical $J(T)$ of T .*

Proof. T is artinian since it is f.d. over k . In this case we know $J(T)$ is the largest nilpotent ideals in T . So $J(T) = \{A \in M_n(k) : a_{ij} = 0 \text{ whenever } i \geq j\} = \left\{ \begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix} \right\}$ □

Proposition 4.4. (*Maschke*) *Let G be a finite group and let k be a field. Show that kG is left and right artinian. Show that kG is semiprimitive if and only if $\text{char}(k) \nmid |G|$.*

Proof. kG is f.d. over k so it is artinian. Now if $\text{char}(k) \nmid$, to show kG is semiprimitive it suffices to show kG is semisimple as kG -module by proposition 4.2.

Let $W \subset V = kG$ be a kG -submodule. By the equivalent definition of semisimple module, it is enough to show that it has a complement in V . Let $\pi : V \rightarrow W$ be any k -linear projection. Define $\pi' : V \rightarrow W$ given by $\pi'(v) = \frac{1}{|G|} \sum_{g \in G} g \pi g^{-1}(v)$ (here we used that $|G| \neq 0$ in k). Note that $\pi'|_W = \pi|_W = \text{id}_W$. Also $h \pi' h^{-1}(v) = \pi'(v)$ for all $h \in G$. So π' is G -equivariant and $\pi' \in \text{Hom}_G(V, W)$. So $\ker \pi'$ is a kG -complement of W in V .

If $\text{char}(k) \mid |G|$, we show that kG is not semisimple. Consider $W = \{f \in kG \mid \sum_{g \in G} f(g) = 0\} \subset kG$ which is a codim-1 submodule. Note that $kG/W \cong k$ with trivial action, since the action of G won't affect the sum of coefficients. It suffices to show that every dim-1 subspace W' of V with trivial action is contained in W . Now for $f \in W'$, $g \cdot f = f$ for all $g \in G$, then f is constant with some value $\lambda \in k$. Then $\sum f(g) = \lambda|G| = 0 \implies f \in W$ □

Proposition 4.5. *Let G be a finite p -group and let k be a field of characteristic p . Using that $Z(G) \neq 1$ and induction on $|G|$, show that $J(kG) = \ker(\phi)$ where $\phi : kG \rightarrow k = \text{End}_k(k)$ defines the trivial representation (which is exactly the isomorphism defined in the second part of proposition 4.4, where we define $\ker(\phi) = W$).*

Proof. Since $Z(G) \neq 0$, take $z \in Z(G)$ with order p . Then $(z - 1)^p = z^p - 1 = 0$ since $\text{char}(k) = p$. Now let M be a simple kG -module. Since z is in the center, note that $(z - 1)M$ is a kG -submodule of M , and hence $(z - 1)M = M$ or 0 . But $(z - 1)^p M = 0 \implies (z - 1)M = 0$. So M is a $k_{\langle z \rangle}^G$ -module. Since $|G/\langle z \rangle| < |G|$, by induction every simple $k(G/\langle z \rangle)$ -module is trivial and so every kG -module is trivial (i.e. k with the trivial action). Hence $J(kG) = \text{Ann}_{kG}(k) = \ker \phi$ $\square$

5 Prime and Semiprime Ideals

Suppose that R is a ring. A **prime ideal** P of R is a proper ideal $P \triangleleft R$ such that if $I, J \triangleleft R$, $IJ \subset P$ then $I \subset P$ or $J \subset P$. We say R is a **prime ring** if (0) is a prime ideal. Note this agrees with the usual definition $xy \in P \implies x \in P$ or $y \in P$ of prime ideals when R is commutative. However these two definitions don't agree in general when R is noncommutative.

We say $P \triangleleft R$ **completely prime** if R/P has no non-trivial zero-divisors. We say R is **domain** if (0) is completely prime.

Proposition 5.1. *Every primitive ideal $P = \text{Ann}_R(M)$ for some simple R -module M is prime.*

Proof. Suppose $I, J \triangleleft R$ with $IJ \subset P$. If $JM = 0$ then $J \subset \text{Ann}_R(M) = P$. Otherwise, $0 \neq JM \subset M$ implies $JM = M$ since M is simple. But then $IM = IJM \subset PM = 0$ $\square$

Example 8. We have seen that for a division ring D , $M_n(D)$ is a simple ring, so (0) is a maximal ideal, and particularly prime. But it is clear that $M_n(D)$ is not a domain, so (0) is not completely prime.

We have the following containments for a general ring

$$\text{MaxSpec}(R) \subset \text{Primitive ideals in } R \subset \text{Spec}(R) \supset \text{Completely prime ideals}$$

If R is left artinian, then $J(R)$ is nilpotent and so contained in every prime ideals. We have seen that $R/J(R)$ is semisimple, i.e. a finite direct product of matrix rings over division rings all of which are simple. By analyzing its ideals, we get $\text{MaxSpec}(R) = \text{Spec}(R)$

Proposition 5.2. *Every prime ideal P in a ring R contains a minimal prime ideal.*

This is done by Zorn's lemma

By considering the ring R/I for an ideal I it follows that every such I has a minimal prime P containing it. We call such a P a **minimal prime** over I .

Theorem 5.1. *If R is a left or right noetherian ring then R has only finitely many minimal prime ideals. Moreover there is a finite product of minimal prime ideals (possibly with repeats) that is zero.*

Proof. Suppose we can find prime ideals $P_1, \dots, P_n$ with $P_1 \cdot \dots \cdot P_n = 0$. By replacing each P_j by a minimal prime contained in it, we may assume that all the P_j are minimal. Then, if P is any minimal prime, $P_1 \cdot \dots \cdot P_n \subset P$. Since P is prime, there is some $P_j \subset P$ and by minimality of P , $P = P_j$. Thus every minimal prime is one of the P_j , so there are only finitely many minimal primes.

So suppose for contradiction that no product of prime ideals is 0 and consider the set S of ideals $I \triangleleft R$ that contain no product of prime ideals. Then $(0) \in S$. By noetherianity we take an maximal element $K \in S$. Then K cannot be prime. Suppose $IJ \subset K$ with $I, J \not\subset K$. Since $(I+K)(J+K) \subset IJ+K = K$, we may assume $K \subsetneq I, J$. By maximality of K , $I, J \notin S$, and hence they contain two products of prime ideals respectively. But then the product of these two products is contained in K . Contradiction. $\square$

Definition 5.1. A **semiprime** ideal in a ring R is an intersection of prime ideals. A ring is **semiprime** if 0 is a semiprime ideal.

Since in a commutative ring every nilpotent element lives in every prime ideal, a semiprime ideal must contain all nilpotent elements. However, the example of a matrix ring $M_n(D)$ shows that this is not true in the noncommutative setting.

However, we have the following criterion for semiprime ideals

Theorem 5.2. An ideal I is semiprime if and only if for every $x \in R$ with $xRx \subset I$, $x \in I$

The proof is very clever and fun. Suppose $I = \cap P$ is an intersection of prime ideals and $xRx \subset I$. Then $(RxR)(RxR) = RxRxR \subset P$ and so $RxR \subset P$ since P is prime. This shows that $x \in I$.

Conversely, suppose I has the latter property. We claim that I is the intersection of prime ideals containing I , and it suffices to show that for $x \notin I$, $x \notin P$ with P prime containing I . Let $x = x_0$. By the property, $x \notin I \implies xRx \not\subset I$. So we can take $x_1 \in xRx \setminus I$. We can continue this and get a sequence $x_0, x_1, x_2, \dots$ such that $x_i \in x_{i-1}Rx_{i-1} \setminus I$. In particular, $x_n \notin I$ for all n . Also $x_n \in J \implies x_m \in J$ for $m \geq n$

By Zorn's lemma, we may choose $I \subset P$ maximal subject to $x_n \notin P$ for all n . So it is enough to show that P is prime. Suppose $KJ \subset P$ with $P \subsetneq K, J$ (as above). Then by maximality we have $x_n \in K$ and $x_m \in J$ for some n, m . WLOG let $n = \max\{n, m\}$. Then $x_n \in K \cap J$ and so $x_{n+1} \subset x_nRx_n \subset KJ \subset P$, contradiction.

Corollary 5.1. TFAE:

1. I is semiprime
2. If $J \triangleleft R$ then $J^2 \subset I \implies J \subset I$
3. If J is a one-sided ideal then $J^2 \subset I \implies J \subset I$

Proof. (1) $\implies$ (3): If J is one-sided, take $x \in J$. Then $xRx \subset J^2 \subset I \implies x \in I$ ($xRx = (xR)x = x(Rx)$ depending on J left or right). (3) implies (2) is trivial. Suppose (2) is true. Pick $x \in R$ with $xRx \subset I$. Then $(RxR)(RxR) = RxRxR \subset I$ and hence $RxR \subset I$. So $x \in I$. $\square$

Corollary 5.2. A semiprime ring has no nilpotent one-sided ideals except 0

Proof. Suppose $0 \neq J$ a nilpotent ideal with n minimal such that $J^n = 0$. Then $n \geq 2$ and so $2n-2 \geq n$. Hence $J^{2n-2} = (J^{n-1})^2 \subset 0$. Since 0 is semiprime, corollary 5.1 (2) gives $J^{n-1} = 0$, contradiction. $\square$

The **prime radical** $N(R)$ of a ring R is the intersection of all the prime ideals of R , equivalently the intersection of all the minimal prime ideals of R .

Note that $N(R)$ is a proper ideal of R unless R is the zero ring since every nonzero ring has a maximal, and hence prime, ideal. A ring is semiprime precisely when the prime radical is zero, since $I^2 = 0 \implies I^2 \subset P$ for all prime $P \implies I \subset P \implies I \subset N(R) = 0 \implies I = 0$.

Theorem 5.3. Suppose that R is a left noetherian ring. Then $N(R)$ is a nilpotent two-sided ideal containing all one-sided nilpotent ideals.

Note that $R/N(R)$ is semiprime and hence contains no non-zero nilpotent one-sided ideals by corollary 5.2. Thus $N(R)$ contains all nilpotent one-sided ideals of R . Since A is left noetherian, theorem 5.1 gives $P_1 \cdot \dots \cdot P_n = 0$ with prime ideals P_i . Since $N \subset P_i$, we get $N^n = 0$, i.e. nilpotent.

6 Essential Extensions and Injective Hulls

We say an R -module E is **injective** if $\text{Hom}_R(-, E)$ is an exact functor. Equivalently, E is injective for $L \hookrightarrow M$ an inclusion, every $\alpha \in \text{Hom}_R(L, E)$ extends to a $\beta \in \text{Hom}_R(M, E)$. For injectivity we have the **Baer's criterion**.

Theorem 6.1. (*Baer's Criterion for injectivity*) *A (left) R -module E is injective if and only if for every left ideal I in R every R -module map $I \rightarrow E$ can be extended to an R -module map $R \rightarrow E$.*

Proof. It suffices to show the if part. Let L be a submodule of M and $\alpha : L \rightarrow E$. Consider the set of all R -linear extensions (L', α') with $L \subset L' \subset M$ and $\alpha' \in \text{Hom}_R(L', E)$ such that $\alpha'|_L = \alpha$ and make it a partially ordered set by $(L', \alpha') \leq (L'', \alpha'')$ if $L' \subset L''$ and $\alpha' = \alpha''|_{L'}$. It is easy to check that the conditions of Zorn's lemma hold, and so this set has a maximal element, say (N, β) . Need to show $N = M$.

Suppose $m \in M \setminus N$. Consider $I = \{r \in R \mid rm \in N\}$ an left ideal and $f : I \xrightarrow{(-) \cdot m} N \rightarrow E$. Then by assumption it extends to $g : R \rightarrow E$. Now define $\beta' : N + Rm \rightarrow E$ by $\beta'(n + rm) = \beta(n) + g(m)$. This is well-defined because for $rm = n \in N \cap Rm$, $r \in I$ and hence $\beta(n) = f(r)$. Thus we have extended β , a contradiction. $\square$

A (left) R -module M is **divisible** if for all $r \in R$, $M \xrightarrow{m \mapsto rm} M$ is surjective.

Injective modules can be hard to pin down in general but we do have the following result for (commutative) principal ideal domains using Baer's criterion.

Corollary 6.1. *If A is a principal ideal domain, then an A -module M is injective if and only if M is divisible. In particular $\mathbb{Q}$ and $\mathbb{Q}/\mathbb{Z}$ are injective as $\mathbb{Z}$ -modules.*

Proof. By Baer's criterion, M is injective if and only if every $f : (r) \rightarrow M$ can be extended to $g : R \rightarrow M$. If M is injective, then for every $m \in M$ and $r \in A$, let $f : (r) \rightarrow M, r \mapsto m$. Then for the extending map g we have $m = f(r) = g(r) = rg(1)$. So M is divisible. Conversely, if M is divisible, for every $f : (r) \rightarrow M$ we define $g : R \rightarrow M$ by letting $g(1) \in M$ such that $rg(1) = f(r)$. $\square$

Remark 6.1. It is easy to see, by the same argument, that injective R -modules are always divisible but the converse is not true in general.

Next we want to prove that the category of R -modules has enough injectives. We do it first for $\mathbb{Z}$ -modules. Given $\mathbb{Z}$ -module A , consider the $\mathbb{Z}$ -module $E(A) = \{f : \text{Hom}_{\mathbb{Z}}(A, \mathbb{Q}/\mathbb{Z}) \rightarrow \mathbb{Q}/\mathbb{Z}\}$. Then $E(A)$ is an injective $\mathbb{Z}$ -module because it is divisible.

Lemma 6.1. *The canonical map $e_A : A \rightarrow E(A)$ defined by $e_A(a)(f) = f(a)$ is an injective $\mathbb{Z}$ -module homomorphism.*

Then proof uses the injectivity of $\mathbb{Q}/\mathbb{Z}$. So the category of $\mathbb{Z}$ -modules has enough injectives.

If M and N are R -modules, a **monomorphism** $M \rightarrow N$ is an injective R -linear homomorphism.

Corollary 6.2. *If R is a ring and E is an injective $\mathbb{Z}$ -module then $\text{Hom}_{\mathbb{Z}}(R, E)$ is an injective R -module under $R \times \text{Hom}_{\mathbb{Z}}(R, E) \rightarrow \text{Hom}_{\mathbb{Z}}(R, E); (a \cdot f)(r) = f(ra)$*

Now given an R -module M , by lemma 6.1 there is an injective $\mathbb{Z}$ -module E containing M viewed as a $\mathbb{Z}$ -module. Then $e : M \rightarrow \text{Hom}_{\mathbb{Z}}(R, E)$ given by $e(m)(r) = rm$ is a monomorphism since $e(m)0$ implies $m = e(m)(1) = 0$

Corollary 6.3. *An R -module M is injective if and only if M is a direct summand of every R -module that contains it.*

Proof. If M is injective and $i : M \hookrightarrow L$, then $M \xrightarrow{id} M$ extends to $f : L \rightarrow M$. Hence $f \circ i = id_M$ and so the extension of M and L/M by L splits, and $L \cong M \oplus L/M$.

Conversely, if M is a direct summand of every module that contains it then it is a direct summand of an injective module $E \cong M \oplus N$ by the observation after corollary 6.2. Then for any $f \in Hom_R(I, M)$, it extends to $f' \in Hom_R(R, E)$. Compose it with the projection $E \rightarrow M$, we get an extension of f in $Hom_R(R, M)$ $\square$

There is a strong relationship between the notion of injective modules and the notion of **essential submodules/extensions**.

Definition 6.1. An **essential submodule** of an R -module M is any submodule N such that $N \cap L \neq 0$ for every nonzero R -submodule L of M . We write $N \leq_e M$ and also say M is an **essential extension** of N . An essential extension $N \leq_e M$ is proper if $N < M$. If $M = R$ as a left module we call N an **essential left ideal**.

Remark 6.2. We can think of essential submodules as dense open sets.

Example 9. An injective module has no proper essential extensions. Indeed, corollary 6.3 implies that it has a direct sum complement in any extension, so it is not essential.

Example 10. If M is a semisimple R -module then M has no essential proper submodules. In lemma 4.1 we have shown that any proper submodule of semisimple M has a direct sum complement in M by Zorn's lemma, and so it is not essential. Conversely, if M is not semisimple we can use Zorn's lemma to get a maximal direct sum of submodules in M , which is an essential submodule

Notice that $N \leq_e L$, $L \leq_e M$ then $N \leq_e M$. If $N_i \leq_e M_i$ for $i \in I$, then $\bigoplus_{i \in I} N_i \leq_e \bigoplus_{i \in I} M_i$.

Proposition 6.1. Suppose that M is an R -module and $N_i \leq_e M_i \leq_e M$ for $i \in I$. If $\sum_{i \in I} N_i$ is a direct sum, then $\sum_{i \in I} M_i$ is a direct sum.

We can reduce it to the case where I is finite and proceed by induction on $|I|$. Here we will use the fact that $L_1 \leq_e N_1$, $L_2 \leq_e N_2$, then $L_1 \cap L_2 \leq_e N_1 \cap N_2$

Proposition 6.2. Suppose that $N \leq M$ are R -modules and $L \leq M$ is maximal subject to $L \cap N = 0$. Then $L \oplus N \leq_e M$ and $(L \oplus N)/L \leq_e M/L$

Proof. If nonzero $K \cap (L \oplus N) = 0$ then $N \cap (L \oplus K) = 0$, contradicts maximality.

If K/L nonzero submodule of M/L then $L \subsetneq K$ and hence $N \cap K \neq 0$ by maximality of L . Thus $L \subsetneq (L \oplus N) \cap K$ and hence $(L \oplus N)/L \cap K/L \neq 0$. $\square$

Corollary 6.4. An R -module M is injective if and only if it has no proper essential extensions.

We have seen that an injective module has no proper essential extension. So suppose that M is non-injective. Then we can find an R -module N containing M such that M is not a direct summand of N . By Zorn's lemma we can find $L \leq N$ maximal subject to $L \cap M = 0$. Then $L \oplus M \leq_e N$ by proposition 6.2. Moreover $(L \oplus M)/L \leq_e N/L$. But we can identify M and $(L \oplus M)/L$ with the natural map, and so N/L is an essential extension of M .

Definition 6.2. Suppose $N \leq M$ are R -modules. We say that N is **essentially closed in M** if $N \leq_e L \leq M$ implies that $N = L$, i.e. N has no proper essential extensions in M .

Proposition 6.3. Suppose that E is an injective R -module and $M \leq E$. Then M is injective if and only if M is essentially closed in E .

The forward direction follows from corollary 6.4. Now suppose M is essentially closed in E . It suffices to show that M has no proper essential extensions. Suppose $M \leq_e N$. Since E is injective, the inclusion $M \rightarrow E$ extends to $f : N \rightarrow E$. Since $\ker(f) \cap M = 0$ and $M \leq_e N$, $\ker(f) = 0$ and so $f : N \rightarrow f(N)$ is an isomorphism and hence $M \leq_e f(N) \leq E$, which implies that $M = f(N)$ since M is essentially closed and $M = N$ since f is injective.

Definition 6.3. An *injective hull* of an R -module M is an injective module that is an essential extension of M , denoted by $E(M)$

The proof of proposition 6.3 shows that any proper essential extension of M can be embedded into $E(M)$, so we can always regard it as a proper essential extension in E . So we can see the injective hull as the maximal essential extension of M . In the meanwhile, it is also the minimal injective modules containing M as we are about to see.

Example 11. $\mathbb{Q}$ is an injective hull of $\mathbb{Z}$ as a $\mathbb{Z}$ -module.

Example 12. if A is a commutative domain then its field of fractions $Q(A)$ is an injective hull of A as an A -module. $Q(A)$ injective is because it is divisible and it is clear an essential extension of A .

Theorem 6.2. (*Existence of Injective Hulls*) Let M be an R -module, then any injective module containing M contains an injective hull E of M . Moreover

1. If $M \leq_e N$ then id_M extends to a monomorphism $N \rightarrow E$.
2. If $M \leq F$ with F injective then id_M extends to a monomorphism $E \rightarrow F$.

Proof. Consider any injective module F containing M and let S be the set of proper essential extension of M in F . We can show that S is chain complete and hence by Zorn's lemma has a maximal element E . WTS E is injective. By proposition 6.3 it is enough to show that E is essentially closed in F . Let E' be any proper essential extension of E in F . Then by transitivity, E' is an essential extension of M in F , contradicts the maximality of E . So E is injective and hence the injective hull of M .

We have shown before that if $M \leq_e N$ id_M extends to a monomorphism $N \rightarrow E$. Now if $M \leq F$ with F injective, id_M extends to a monomorphism $E \rightarrow F$ since $M \leq_e E$. So E can be embedded into any injective module containing M . $\square$

Proposition 6.4. (*Uniqueness of Injective Hulls*) M and N are R -modules with injective hulls E and F respectively then any isomorphism $f : M \rightarrow N$ extends to an isomorphism $E \xrightarrow{\sim} F$. In particular if $M = N$ then id_M extends to an isomorphism $E \xrightarrow{\sim} F$.

Proof. Since f can be viewed as a monomorphism $f : M \rightarrow N \hookrightarrow F$, it extends to a monomorphism $f' : E \rightarrow F$ since F is injective and $M \leq_e E$. So it remains to show that f' is surjective. But $N = f(M) = f'(M) \leq f'(E)$. So $f'(E) \leq_e F$ and $f'(E)$ is injective and hence essentially closed in F by proposition 6.3. So $f'(E) = F$ $\square$

Since $E(M)$ is only unique up to (possibly non-unique) isomorphism a true assertion about $E(M)$ must be true about every choice of $E(M)$.

Example 13. If M_1 and M_2 are R -module then $E(M_1 \oplus M_2) = E(M_1) \oplus E(M_2)$. If $M \leq N$, we have seen that this extends to a monomorphism $E(M) \hookrightarrow E(N)$. If moreover $M \leq_e N$, this is in fact an isomorphism, because $M \leq_e N \leq_e E(N) \implies M \leq_e E(N) \implies E(M) \leq_e E(N)$

6.1 Exercise

Proposition 6.5. *Show that if $(E_i)_{i \in I}$ is any family of R -modules then $\prod_{i \in I} E_i$ is an injective R -module if and only if each E_i is an injective R -module. Deduce that finite direct sums of injective modules are always injective and direct summands of injective modules are injective.*

This is seen by compositing with the inclusion $i_i : E_i \hookrightarrow \prod E_j$ and projection $\pi_i : \prod E_j \rightarrow E_i$

Proposition 6.6. *Suppose M is a left R -module. Then M is semisimple if and only if M has no proper essential submodule*

Proof. We have seen that any proper submodule of a semisimple module has a direct sum complement and so is not essential. Conversely, for $N \subset M$, take L maximal subject to $L \cap N = \emptyset$. Then $L \oplus N \leq_e M$ and hence $L \oplus N = M$ $\square$

Proposition 6.7. *Suppose that A is a commutative domain. Show that its field of fractions $Q(A)$ is an injective hull of A viewed as a (left) A -module.*

Proof. It is clear that $A \leq_e Q(A)$. To show $Q(A)$ is an injective A -module, we use Baer's criterion. For an ideal I and $\phi : I \rightarrow Q(A)$, we define $\varphi : A \rightarrow Q(A)$ given by $1 \mapsto \phi(r)/r$ for any $0 \neq r \in I$. This is well defined because $r\phi(r') = \phi(rr') = r'\phi(r)$ for $r, r' \in I$ $\square$

Proposition 6.8. *Suppose $N < M$ are R -modules show that $E(M)$ is isomorphic to a submodule of $E(N) \oplus E(M/N)$.*

Proof. $N \hookrightarrow M \hookrightarrow E(M) \implies E(N) \hookrightarrow E(M)$ since $E(M)$ is injective and $N \leq_e E(N)$. Since $E(N)$ is injective, it can be identified with a direct summand of $E(M)$. So we get a projection $\alpha : E(M) \rightarrow E(N)$.

$M \rightarrow M/N \rightarrow E(M/N) \implies \beta : E(M) \rightarrow E(M/N)$. Now we have $E(M) \xrightarrow{(\alpha, \beta)} E(N) \oplus E(M/N)$. NTS (α, β) is injective. Note that $\ker \beta \cap M = N$ and $\alpha|_{E(N)} = \text{id}_{E(N)}$ and so $\ker \alpha \cap E(N) = 0$. So $\ker(\alpha, \beta) \cap M = \ker \alpha \cap \ker \beta \cap M = 0 \implies \ker(\alpha, \beta) = 0$ since $M \leq_e E(M)$, so done. $\square$

7 Goldie's Rank

Lemma 7.1. *If A is a torsionfree $\mathbb{Z}$ -module then $E(A) \cong \mathbb{Q} \otimes_{\mathbb{Z}} A$.*

We can see $\mathbb{Q} \otimes_{\mathbb{Z}} A$ as the localization of A w.r.t. $S = \mathbb{Z} \setminus \{0\}$. Then A_S is divisible and hence injective. A can be identified as a submodule of A_S because A is torsion-free, and is clearly essential in A_S

There is a classical notion of the rank of a torsion-free abelian group A that corresponds to the dimension of $\mathbb{Q} \otimes_{\mathbb{Z}} A$. We'd like to extend this to more general R -modules using injective hulls.

Definition 7.1. *An R -module M is indecomposable if it cannot be written as a direct sum $M = L \oplus N$ with L and N both nonzero.*

We'd like to define the rank of an R -module in terms as the number of summands when you decompose its injective hull as a direct sum of indecomposable injective modules. This would generalise the notion of the rank of a torsion-free abelian group, the dimension of a vector space and the function on semisimple artinian modules $\oplus_{i \in I} M_i$ that counts the number of copies of simple modules M_i .

There are two difficulties: first an injective module may not decompose as a direct sum of indecomposable modules and second the number of factors in the sum might depend on the choice of decomposition. Of course these are already issues for defining the dimension of a vector space.

Definition 7.2. An R -module M has **finite (Goldie) rank** if $E(M)$ is a finite direct sum of indecomposable injective modules. If $E(M) = \bigoplus_{i=1}^n E_i$ with E_i indecomposable injective, we call n the Goldie rank of M and write $\text{rank}(M) = n$

This leads us naturally to the following notion.

Definition 7.3. An R -module M is **uniform** if it is nonzero and the intersection of any two nonzero submodules of M is nonzero; equivalently if every nonzero submodule of M is an essential submodule of M .

Remark 7.1. We can think of this as irreducible topological space. Then any open subset is dense.

Remark 7.2. In example sheet we showed that a left noetherian domain is uniform and their regular elements form a left Ore set (Should have included this in the notes. Considering the ascending chain $Rr \subset Rr + Rrs \subset \dots$)

It is easy to see that every nonzero submodule of a uniform module is uniform.

Lemma 7.2. A nonzero R -module M is uniform if and only if $E(M)$ is indecomposable.

Proof. Suppose M is uniform and $E(M) = L \oplus N$. Then $(L \cap M) \cap (N \cap M) \subset L \cap N = 0$ implies that $L \cap M = 0$ or $N \cap M = 0$ since M is uniform. It follows that $L = 0$ or $N = 0$ since $M \leq_e E(M)$.

Conversely, suppose M is not uniform. Then we can find $L, N \leq M$ with $L \cap N = 0$. Then inclusion $N \hookrightarrow M$ induces a monomorphism $E(N) \hookrightarrow E(M)$, so we can regard $E(N)$ as a submodule of $E(M)$. Now $E(N) \cap L = 0$ since $L \cap N = 0$ and $N \leq_e E(N)$. Then $E(N)$ is a proper injective submodule of $E(M)$. Then by corollary 6.3 it is a direct summand of $E(M)$, and hence $E(M)$ is decomposable. $\square$

It follows that a nonzero injective module is uniform if and only if it is indecomposable.

Proposition 7.1. An R -module M has finite rank if and only if it has an essential submodule that is a finite direct sum of uniform modules.

Proof. Suppose $\bigoplus_{i \in I} M_i \leq_e M$ for uniform modules M_i and I finite. Then $\bigoplus_{i \in I} M_i \leq_e E(M)$ by transitivity and so $E(M) = E(\bigoplus_{i \in I} M_i) = \bigoplus_{i \in I} E(M_i)$. Then $E(M_i)$'s are indecomposable by lemma 7.2, so M has finite rank.

Conversely, if M has finite rank then $E(M) = \bigoplus_{i \in I} E_i$ with I finite E_i uniform injective because E_i is a direct summand of injective module and indecomposable. Let $M_i = E_i \cap M \neq 0$ since $M \leq_e E(M)$. It is also uniform since E_i is so. Since $M_i \leq_e E_i$, we have $\bigoplus_{i \in I} M_i \leq_e \bigoplus_{i \in I} E_i = E(M)$. Hence $\bigoplus_{i \in I} M_i \leq_e M$. $\square$

Lemma 7.3. If an R -module M is a direct sum of n uniform submodules then it contains no direct sum of $n + 1$ nonzero submodules

This is done by induction on n . From this we see that rank of M does not depend on the choice of decomposition of $E(M)$.

It is easy to see that $\text{rank}(\bigoplus_{i \in I} M_i) = \sum_{i \in I} \text{rank}(M_i)$ if they are finite.

Proposition 7.2. Let M be an R -module and $n \geq 0$. The following are equivalent

1. $\text{rank}(M) = n$
2. M has an essential submodule which is a direct sum of n uniform submodules
3. M contains a direct sum of n nonzero submodules but no direct sum of $n + 1$ nonzero submodules

(1) implies (3) uses last lemma. The rest of the proof is pretty standard.

Corollary 7.1. *Suppose $N \leq M$ R -modules. Then*

1. *If M has finite rank then $\text{rank} N \leq \text{rank} M$.*
2. *If N has finite rank then $\text{rank} N = \text{rank} M$ if and only if $N \leq_e M$*
3. *If N and M/N both have finite rank then $\text{rank} M \leq \text{rank} N + \text{rank} M/N$.*

(1) and (2) follow from last proposition. For (3), Let $L \leq M$ be maximal subject to $L \cap N = 0$. Then L is isomorphic to a submodule of M/N . Then by (1), $\text{rank} L \leq \text{rank} M/N$. Since $L \oplus N \leq_e M$, (2) gives $\text{rank} M = \text{rank} L + \text{rank} N \leq \text{rank} N + \text{rank} M/N$.

Corollary 7.2. *If M is an R -module of finite rank then for every monomorphism $f : M \rightarrow M$, $f(M) \leq_e M$.*

Since $f(M) \cong M$, $\text{rank} f(M) = \text{rank} M$. And so this is by corollary 7.1 (2).

Corollary 7.3. *If R has finite rank as a left R -module and $x \in R$ is not a right zerodivisor, i.e. $rx = 0$ for $r \in R$ implies $r = 0$, then $Rx \leq_e R$ and for all $a \in R$, $(Rx)a^{-1} = \{r \in R : ra \in Rx\} \leq_e R$.*

Notice the map $R \xrightarrow{(-) \cdot x} Rx$ is a monomorphism.

Theorem 7.1. (Goldie). *An R -module M has finite rank if and only if M does not contain an infinite direct sum of nonzero submodules*

Proof. The forward implication follows from proposition 7.2 (3). Conversely, suppose M is not of finite rank. Then $E(M)$ is not a finite direct sum of indecomposable injective modules. So we can write $E(M) = X_1 \oplus Y_1$ for some nonzero injective modules X_1, Y_1 , one of which is not a finite direct sum of indecomposable injective modules. WLOG assume it is X_1 . Then we can write $X_1 = X_2 \oplus Y_2$. Repeat this argument and we can get sequence X_n and Y_n of injective nonzero modules such that $X_{i-1} = X_i \oplus Y_i$, and X_n is not a finite sum of indecomposable injective modules for all n .

Note that $Y_n \cap (\oplus_{i \geq n+1} Y_i) \leq Y_n \cap X_n = 0$ for all n . Since $M \leq_e E(M)$, $M \cap Y_n \neq 0$ for all n and hence $\sum_n M \cap Y_n$ is a direct sum contained in M . Thus M contains an infinite direct sum of nonzero submodules. $\square$

Corollary 7.4. *If M is a noetherian R -module then M has finite rank.*

This follows from theorem 7.1 because an infinite direct sum of nonzero submodules produces a strictly ascending chain in M (and also a strictly descending chain) as we have seen before.

Theorem 7.2. *A ring R is left noetherian if and only if every direct sum of injective (left) R -modules is injective.*

The proof uses finiteness. Let $E = \oplus_{i \in I} E_i$ direct sum of injective modules. Since any ideal I is finitely generated, any $f \in \text{Hom}_R(I, E)$ lands in only finitely many E_i , and their direct sum (which is finite) is an injective modules, so f extends to R . Conversely, we can take an ascending chain $I_1 \leq I_2 \leq \dots$ and consider $E = \oplus_{i \geq 1} E(R/I_i)$

Corollary 7.5. *If R is left noetherian then every injective (left) R -module is a direct sum of uniform injective R -modules.*

Let E be an injective R -module. We use Zorn's lemma to get a maximal family $\{M_i\}$ of finitely generated submodules of E such that the sum $\sum_i M_i$ is direct. By maximality, $\oplus_i M_i \leq_e E$. Hence $\oplus_i E(M_i) \leq_e E$ and is injective by theorem 7.2. So $\oplus_i E(M_i) = E$. Now each M_i is noetherian and so has finite rank by corollary 7.4. Thus each $E(M_i)$ is a finite direct sum of uniform injective modules and E is a direct sum of uniform injective modules.

Lemma 7.4. *Suppose that U is a uniform R -module over a left noetherian ring R . There is a unique prime ideal P of R such that P is the annihilator of a nonzero submodule of U and P contains the annihilators of all non-zero submodules of U . We call P the **assassinator** of U and write $ass_R(U)$*

Proof. Since R is left noetherian we may find $M \leq U$ such that $P = Ann_R(M)$ is maximal amongst the annihilators of nonzero submodules of U . If $N \leq U$ another nonzero submodule then $N \cap M \neq 0$ since U is uniform. Then $P = Ann_R(M \cap N) \geq Ann_R(N)$ by maximality of P . So it is enough to show that P is prime.

If $I, J \triangleleft R$ with $IJ \subset P$, then $IJM = 0$. We have $JM = 0$ and $J \subset P$, or $0 \neq JM \leq U$ and so $I \subset Ann_R(JM) \subset P$ $\square$

Remark 7.3. Similar to the proof of the existence of associated primes.

Lemma 7.5. *Let P be a prime ideal in a left noetherian ring R , and let $U = I/P$ be a uniform left R -submodule of R/P . Then $E(U)$ is a uniform injective R -module with assassinator P .*

Proof. Since U is uniform, by lemma 7.2 $E(U)$ is indecomposable and hence uniform and $ass_R(U) = ass_R(E(U))$ (not sure why). Also $P \leq ass_R(U)$ since $PU = 0$. Now any nonzero submodule of U is of the form J/P with $P < J \triangleleft R$. Let $ann_R(J/P) = K$. Then $K(J/P) = 0$ gives $KJ \subset P$ and hence $K \subset P$ since P is prime. $\square$

Proposition 7.3. *Let A be a commutative noetherian ring then $P \mapsto E(A/P)$ defines a bijection between $Spec A$ and the set of isomorphism classes of uniform injective A -modules.*

First notice that A/P is uniform. This is because for $a, b \neq 0$ in A/P , $a, b \notin P \implies ab \notin P$. So $ab \neq 0$ in A/P and hence $0 \neq (ab) \subset (a) \cap (b)$. So $E(A/P)$ is uniform has the assassinator P by lemma 7.5. This shows that $E(A/P) \not\cong E(A/P')$ for $P \neq P'$ and so the map in the statement is a well-defined injection.

Now suppose E is any uniform injective module and let P be the assassinator of E . $P = Ann_A(M)$ for some $0 \neq M \leq E$. Take $0 \neq x \in M$. Then $P = Ann_A(M) \leq ann_A(x) = Ann_A(Ax) \implies P = Ann_A(Ax)$ since A is commutative and P is maximal. Hence $Ax \cong A/P$ and so $E(Ax) \cong E(A/P)$ is isomorphic to a submodule of E . Since E is uniform, $E(Ax) \leq_e E$ and hence $E(A/P) = E(Ax) = E$ since $E(Ax)$ is injective. So the map in the statement is surjective.

Therefore, the correct analogue of $Spec A$ for a noncommutative (left) noetherian ring is the collection of isomorphism classes of uniform injective modules.

7.1 Exercise

Proposition 7.4. *Suppose M is a R -module of finite rank and $N \leq M$ is essentially closed. Show that $rank M/N$ is finite and $rank M = rank N + rank M/N$.*

Proof. Let $L \leq M$ maximal subject to $L \cap N = 0$. It suffices to show that $L \oplus N/N \leq_e M/N$, since $L \cong L \oplus N/N$ and then $rank(L) = rank(M/N)$. Suppose there is $M'/N \subset M/N$ such that $L \oplus N/N \cap M'/N = 0$. Then $N \leq M'$ and $L \oplus N \cap M' \subset N \implies M' \cap L = 0 \implies N \leq_e M' \leq M \implies N = M'$, since N is essentially closed in M . So $M'/N = 0$ $\square$

Proposition 7.5. *Classify all uniform finitely generated $\mathbb{Z}$ -modules. Show that a finitely generated uniform $\mathbb{Z}$ -module may have a quotient of arbitrarily large finite rank. By considering $\mathbb{Q}/\mathbb{Z}$ find an explicit description of all uniform injective $\mathbb{Z}$ -modules*

Since such modules can't be written as a product, only $\mathbb{Z}$ and $\mathbb{Z}/p^r\mathbb{Z}$ satisfy the condition for p prime. Also $\mathbb{Z}/p_1 \dots p_n\mathbb{Z}$ has rank n for distinct primes p_i since each $\mathbb{Z}/p_i\mathbb{Z}$ is uniform. Now $\mathbb{Q}$ is a uniform injective $\mathbb{Z}$ -module. Since for $N \leq_e M$, N uniform implies M uniform, we need to find $E(\mathbb{Z}/p\mathbb{Z})$. $\mathbb{Q}/\mathbb{Z}$ is injective, so if we localize at (p) , $\mathbb{Q}_{(p)}/\mathbb{Z}_{(p)}$ is injective and $\frac{1}{p}\mathbb{Z}_{(p)}/\mathbb{Z}_{(p)} \subset \mathbb{Q}_{(p)}/\mathbb{Z}_{(p)}$ is the injective hull

8 Torsion and Noncommutative Localisation

Recall that a **multiplicatively closed (m.c.)** subset of a ring R is a subset S of R containing 1 and closed under multiplication in the ring.

Given a m.c. subset S of R and M a (left) R -module let $t_S(M) = \{m \in M \mid sm = 0 \text{ for some } s \in S\}$.

Definition 8.1. *We say M is S -torsion-free if $t_S(M) = 0$ and S -torsion if $t_S(M) = M$.*

We want to make $t_S(M)$ is an R -module by giving more restrictions on S . To begin with, we consider the R -module R/Rs for each $s \in S$. It is clear that $1 + Rs \in t_S(R/Rs)$. Since it is an R -module, we have $t_S(R/Rs) = R/Rs$. This requires that for all $a \in R$, there exists $t \in S$ and $r \in R$ such that $ta = rs$. This gives the following definition:

Definition 8.2. *We say that a m.c. subset S of R satisfies the **left Ore condition** if for every $s \in S$ and $a \in R$ there exist $t \in S$ and $b \in R$ such that $ta = bs$. We call such a set a **left Ore set**. An **Ore set** in R is a m.c. subset that is both a left Ore set and a right Ore set.*

Of course in a commutative ring R every m.c. subset of R is an Ore set. More generally, any m.c. subset of $Z(R)$ is an Ore set.

Remark 8.1. If S is a left Ore set and $s_1, \dots, s_n \in S$ then we can show inductively that there exist $a_1, \dots, a_n \in R$ such that $a_1 s_1 = \dots = a_n s_n$

Lemma 8.1. *Suppose that S is a left Ore set in R . Then for any (left) R -module M , $t_S(M)$ is an R -submodule of M .*

Proof. For m_1, m_2 in $t_S(M)$, suppose $s_1 m_1 = s_2 m_2 = 0$. Then the remark, there exist $a_1, a_2 \in R$ such that $a_1 s_1 = a_2 s_2 = b$. Hence $b(m_1 + m_2) = 0$ and $m_1 + m_2 \in t_S(M)$. For $a \in R$, the left Ore condition implies that there exist $t \in S, c \in R$ such that $ta = cs_1$. Hence $tam_1 = cs_1 m_1 = 0$ and so $am_1 \in t_S(M)$ $\square$

If S is a left Ore set in R and M is a left R -module. The submodule $t_S(M)$ is called the S -torsion submodule of M .

Remark 8.2. It is clear that $M/t_S(M)$ is S -torsionfree

Definition 8.3. *We call $a \in R$ a regular element if a is not a zero-divisor i.e. if $b \in R$ with $ab = 0$ or $ba = 0$ then $b = 0$.*

Let R be a ring and S a m.c. subset of R consisting of regular elements. Recall that if R is commutative then there is a localisation R_S of R at S such that every element of R_S can be written as r/s with $r \in R$ and $s \in S$.

Definition 8.4. A ring T equipped with an injective ring homomorphism $\phi : R \rightarrow T$ is a **left ring of fractions** of R with respect to S if $\phi(S) \subset T^\times$ and every element of T can be written as $\phi(s)^{-1}\phi(a)$ for some $a \in R$ and $s \in S$

By identifying R with its image, we can think of T as R_S as in the commutative case.

Lemma 8.2. By adding the condition that S consists of regular elements in above settings, we have

1. S is a left Ore set
2. $\phi(s)^{-1}\phi(a) = \phi(t)^{-1}\phi(b)$ if and only if there exist $c, d \in R$ such that $ca = db$, $cs = dt \in S$
3. Given $t_1, \dots, t_n \in T$ there are $a_1, \dots, a_n \in R$ and $s \in S$ such that $t_i = \phi(s)^{-1}\phi(a_i)$

Remark 8.3. (2) and (3) are analogous to the feature of fractions we are familiar with. For instance, (3) is just "finding common denominator"

Proof. (1) Suppose $s \in S$ and $a \in R$. Since $\phi(s) \in T^\times$, we have $\phi(a)\phi(s)^{-1} \in T$, and so of the form $\phi(a)\phi(s)^{-1} = \phi(t)^{-1}\phi(b)$ for some $t \in S$ and $b \in R$. Rearrange to get $\phi(ta) = \phi(bs)$. Then $ta = bs$ since ϕ injective.

The proofs of (2) and (3) are nothing special. □

Proposition 8.1. (Universal Property for left localisation). Let R be a ring and $S \subset R$ a left Ore set consisting of regular elements and let $\phi : R \rightarrow T$ make T a left ring of fractions of R with respect to S . Then for every ring homomorphism $\theta : R \rightarrow U$ such that $\theta(S) \subset U^\times$ there is a unique ring homomorphism $\varphi : T \rightarrow U$ such that $\varphi \circ \phi = \theta$.

The construction is not interesting, just as every universal property argument. We only concern that every such θ factors through T .

Corollary 8.1. The left rings of fractions for R with respect to S is unique up to isomorphism.

Next we discuss the condition for a left ring of fraction of R with respect to S to exist.

Proposition 8.2. (Asano). Let R be a ring and $S \subset R$ a m.c. subset consisting of regular elements then R has a left ring of fractions with respect to S if and only if S is a left Ore set.

The forward direction is already shown. We will have to omit the proof of the other half. So to make (T, ϕ) exist, it suffices to make S consisting of regular elements a left Ore set.

If R is a ring and $S \subset R$ is a left Ore set consisting of regular elements we'll write $S^{-1}R$ to denote a left ring of fractions of R with respect to S . We will routinely identify R with its image in $S^{-1}R$

Proposition 8.3. If $S \subset R$ is an Ore set consisting of regular elements then $S^{-1}R = RS^{-1}$.

This follows from the right version of Ore condition and universal property.

Remark 8.4. If R is a left noetherian ring and S is a left Ore set consisting of regular elements then $S^{-1}R$ is a left noetherian ring by considering the contraction of ideals as in the commutative case.

Definition 8.5. A **classical left quotient ring** for a ring R is a left ring of fractions with respect to the set of all regular elements of R .

We've seen that R has a classical left quotient ring if and only if the set of regular elements is a left Ore set. Note that if R is a domain then a left/right classical ring of quotients is a division ring.

Recall the definition of **left annihilator** $lann_R(X)$ and **right annihilator** $rann_R(X)$ of a set $X \subset R$. Note that if $I = lann_R(X)$ then $X \subset rann_R(I)$ and hence $I = lann_R(rann_R(I))$

Proposition 8.4. *Suppose that R has a classical left quotient ring Q that is left noetherian. Then R has finite rank as a left R -module and the set of left annihilators in R has no strict ascending chains. Moreover if Q is semiprimitive and artinian then R is semiprime.*

Proof. Let $I_1 \leq I_2 \leq \dots$ be an ascending chain of left annihilators. Let $J_n = rann_R(I_n)$. Then $J_1 \geq J_2 \geq \dots$ is a descending chain and $I_n = lann_R(J_n)$ by above observations.

Now $lann_Q(J_1) \leq lann_Q(J_2) \leq \dots$ is an ascending chain of left ideals of Q . Since Q is left noetherian, this chain terminates. Moreover $I_n = lann_Q(J_n) \cap R$ and so the chain of I_n also terminates and we have the a.c.c. on the set of left annihilators in R as claimed.

Now suppose R is not of finite rank as a left R -module. By theorem 7.1, it contains an direct sum of infinitely many nonzero left ideals, say $\oplus_{i \geq 1} I_i$. Take any $a_n \in I_n \setminus 0$. By corollary 7.4, Q is left noetherian implies Q is of finite rank, and so the sum $\sum_{i \geq 1} Qa_i$ is not direct. Thus there are $q_1, \dots, q_m \in Q$ such that $\sum_{i=1}^m q_i a_i = 0$ but $q_m a_m \neq 0$. Also by Ore condition there is $x \in R$ a regular element such that $xq_i \in R$ for all i . Then $\sum xq_i a_i = 0$ where $xq_i a_i \in I_i$ are independent and so $x_m q_m a_m = 0$. Since x is a unit in Q , we have $q_m a_m = 0$, a contradiction. So R has finite rank.

Finally assume that Q is semiprimitive and artinian. By corollary 5.1, to show $(0) \triangleleft R$ is semiprime it is enough to show that $N^2 = 0 \implies N = 0$. Let $L = rann_R(N)$. It suffices to show that L has a regular element.

If $I \triangleleft_l R$ then either $I \leq L$ or $0 \neq NI \subset L \cap I$. Thus $L \leq_e R$. We claim $QL \leq_e L$. For $J \triangleleft_l Q$, take $s^{-1}r \in J$ with $s \in R$ regular and $0 \neq r \in R$. Then $r \in J \cap R$. So $R \cap J$ nonzero ideal and hence $0 \neq R \cap J \cap L \subset QL \cap J$

Since Q is semiprimitive and artinian and hence semisimple, $QL = Q$ since it has no essential proper submodules by listing out its ideals. So there are $q_1, \dots, q_n \in Q$ and $l_1, \dots, l_n \in L$ such that $\sum q_i l_i = 1$. As before there is $x \in R$ regular such that $xq_i \in R$ for all i . Thus $x = \sum xq_i l_i \in L$. So L has a regular element. $\square$

Definition 8.6. *A **left Goldie ring** is a ring such that R has finite rank as a left R -module and R has the a.c.c. on left annihilators.*

Every left noetherian ring is a left Goldie ring since noetherian rings have finite rank by corollary 7.4.

8.1 Goldie's Theorem

We've just proven that a ring with a semiprimitive artinian left classical ring of quotients is a semiprime left Goldie ring. Goldie's Theorem tells us that the converse is also true.

To prove this we must show that the set of regular elements in a semiprime left Goldie ring is a left Ore set. That is given any $a, s \in R$ with s regular we must find t regular such that $ta \in Rs$.

When R has finite rank, corollary 7.3 gives $(Rs)a^{-1} = \{r \in R : ra \in Rs\} \leq_e R$. Thus to see the regular elements form a left Ore set it suffices to prove that every essential left ideal contains a regular element. This will be our short-term goal when R is semiprime and left Goldie.

Proposition 8.5. *Let R be a ring and let*

$$J = \{a \in R \mid Ia = 0 \text{ for some } I \leq_e R\} = \{a \in R \mid lann_R(a) \leq_e R\}$$

Then

1. $J \triangleleft R$
2. *If R has the acc. on left annihilators then J is nilpotent.*

3. If R is a semiprime left Goldie ring then $J = 0$.

Proof. (1) For $a, b \in J$, $\text{lann}_R(a) \cap \text{lann}_R(b) \leq_e R$ and $\text{lann}_R(a) \cap \text{lann}_R(b) \leq \text{lann}_R(a+b) \implies a+b \in J$. It remains to prove that if $a \in J$ and $r \in R$ then $ra \in J$. To this end consider $I = \{x \in R \mid xr \in \text{lann}_R(a)\}$. Then $Ira = 0$ and it is enough to show that $I \leq_e R$.

Let $0 \neq L \leq R$. If $Lr = 0$, then $L \leq I$. If $Lr \neq 0$, then $0 \neq Lr \cap \text{lann}_R(a)$ since $\text{lann}_R(a) \leq_e R$ and so there is $x \in L$ such that $xr \neq 0$ and $xra = 0$ and hence $x \in I$. Thus $0 \neq x \in L \cap I$ and we get $I \leq_e R$.

(2) Consider the ascending chain $\text{lann}_R(J) \leq \text{lann}_R(J^2) \dots$. By assumption, there is some k such that $\text{lann}_R(J^k) = \text{lann}_R(J^{k+1})$. We claim that $\text{lann}_R(J^k) = R$ and so $J^k = 0$.

If not, take $x \in R \setminus \text{lann}_R(J^k)$ with $\text{lann}_R(x)$ maximal subject to these elements. This is possible because of acc on left annihilators. For $a \in J$, $\text{lann}_R(a) \cap Rx \neq 0$ since $\text{lann}_R(a) \leq_e R$. Then there is $s \in R$ such that $sxa = 0$ but $sx \neq 0$. This implies that $\text{lann}_R(x) < \text{lann}_R(sx)$ and hence $sx \in \text{lann}_R(J^k)$ by maximality of x . Since this true for all $a \in J$, we have $sx \in \text{lann}_R(J^{k+1}) = \text{lann}_R(J^k)$, contradiction.

(3) follows from (2) since (0) is semiprime and hence $J^k = 0 \implies J = 0$. $\square$

Lemma 8.3. *Let R be a semiprime left Goldie ring. For $a \in R$ the following are equivalent:*

1. a is regular
2. $\text{lann}_R(a) = 0$
3. $Ra \leq_e R$

Proof. (1) implies (2) is clear, and (2) implies (3) is by corollary 7.3 which only uses that R has finite rank. Suppose (3) holds. Note that for $x \in \text{rann}_R(a)$, $Ra \leq \text{lann}_R(x)$ and so $x \in J$ with J in the last proposition. But $J = 0$ by (3) in last proposition. So we have $\text{rann}_R(a) = \text{rann}_R(Ra) = 0$. Let $I = \text{lann}_R(a)$ and let $K \triangleleft_l R$ maximal subject to $I \cap K = 0$ by Zorn's lemma. WTS $K \leq_e R$ and so $I = 0$. It is enough to show that $\text{rank} K = \text{rank} R$

Now $\text{rank} Ra = \text{rank} R$ since $Ra \leq_e R$. Also $\text{rank} K = \text{rank} Ka$ because $K \xrightarrow{a} Ka$ is an isomorphism as $K \cap \text{lann}_R(a) = 0$. So it suffices to show that $Ka \leq_e Ra$.

Suppose $0 \neq ra \in Ra$ and let $L = (I + K)r^{-1} \leq_e R$ because $I + K \leq_e R$ by maximality of K . Then $\text{rann}_R(L) = 0$ by last proposition again. Thus $0 \neq Lra \leq (I + K)a = Ka$, i.e. $Rra \cap Ka \neq 0$. Hence $Ka \leq_e Ra$ and we are done. $\square$

Lemma 8.4. *If R is a semiprime left Goldie ring then R has the dcc. on left annihilators.*

Consider a descending chain $I_1 \geq I_2 \dots$, then $\text{rank} I_i$ is a descending chain in $\mathbb{N}$ and so terminates. This implies that $I_n \leq_e I_m$ for $n \geq m$ and sufficiently large m .

Let $n \geq m$ and $x \in I_m$, then $J = I_n x^{-1} \leq_e R$ because $0 \neq Rrx \cap I_n$ for any $r \in R$ and $JxX_n = 0$ where $I_n = \text{lann}_R(X_n)$. By proposition 8.5 (3) again, $\text{rann}_R(J) = 0$. So $xX_n = 0$ and $x \in I_n$, and hence $I_n = I_m$

Proposition 8.6. *(Goldie's Regular Element Lemma) Let R be a semiprime left Goldie ring and $I \triangleleft_l R$. Then $I \leq_e R$ if and only if I contains a regular element of R .*

Proof. If $a \in I$ is regular, then $Ra \leq I$ and $Ra \leq_e R$ by lemma 8.3. Conversely, suppose $I \leq_e R$. By last lemma, we can take $a \in I$ with $X = \text{lann}_R(a)$ minimal amongst the left annihilators of elements in I . It suffices to show that $Ra \leq_e I$ since then a is regular by lemma 8.3

So we consider $J \leq I$ with $J \cap Ra = 0$ and try to show $J = 0$. Given $x \in J$, $Rx \cap Ra = 0$. So $\text{lann}_R(x + a) = \text{lann}_R(x) \cap \text{lann}_R(a) \leq \text{lann}_R(a)$. By minimality of a , we get $\text{lann}_R(a) \leq \text{lann}_R(x)$ for all $x \in J$. So $XJ = 0$ and hence $(JR \cap X)^2 \subset XJR = 0 \implies JR \cap X = 0$ since R is semiprime.

It follows that $JR \xrightarrow{a} JR$ is a monomorphism and hence $JRa \leq_e JR$. Thus as $J \cap JRa \subset J \cap Ra = 0$, we get $J = 0$ $\square$

Corollary 8.2. *If R is a prime left Goldie ring then every nonzero (two-sided) ideal contains a regular element.*

Indeed, give nonzero $I \triangleleft R$ and $J \triangleleft_l R$, $0 \neq IJ \subset I \cap J$ since (0) is prime. Hence any nonzero two-sided ideal is essential and $I \leq_e R$. Now this follows from proposition 8.6

Theorem 8.1. (Goldie's Theorem) *A ring R has a semiprimitive artinian classical left quotient ring if and only if R is a semiprime left Goldie ring.*

Proof. The forward direction is proposition 8.4. Now suppose R is a semiprime left Goldie ring.

First we show that the set of regular elements form a left Ore set, i.e. the classical left quotient ring exists. Let $a, x \in R$ with a regular. By lemma 8.3 $Ra \leq_e R$ and so $(Ra)x^{-1} \leq_e R$ and Goldie's Regular Element Lemma thus gives a regular element $s \in (Ra)x^{-1}$. So the set of regular elements of R is a left Ore set and has the classical left quotient ring Q .

To prove that Q is semiprimitive artinian it suffices to show that Q is a semisimple Q -module by proposition 4.2, or equivalently Q has no proper essential left ideals. Suppose that I is an essential left ideal of Q . We claim that $I \cap R \leq_e R$ and so contains a regular element which is a unit in Q .

To see this, suppose $0 \neq r \in R$. Then $Qr \cap I \neq 0$ so there is $0 \neq s, a \in R$ with s regular such that $s^{-1}ar \in I$. Hence $0 \neq ar \in Rr \cap (I \cap R)$ and hence $I \cap R \leq_e R$ $\square$

Corollary 8.3. *Every semiprime left noetherian ring has a semiprimitive artinian classical left quotient ring.*

This is because left noetherian rings are left Goldie

Corollary 8.4. *A ring R has a simple artinian classical left quotient ring if and only if R is a prime left Goldie ring.*

Proof. Suppose R is prime left Goldie. With the settings in theorem 8.1, we need to show that Q has no nonzero proper ideals. Let $0 \neq I \triangleleft Q$. By corollary 8.2, $I \cap R$ contains a regular element which is a unit in Q . So $I = Q$. Conversely, suppose $0 \neq I, J \triangleleft R$ with $IJ = 0$, then $0 \neq QI \triangleleft Q \implies QI = Q$ (S consists of regular elements, so $I \neq 0 \implies QI \neq 0$) since Q is simple. Similarly $QJ = Q$. But then $0 = QIJ = QJ = Q$, contradiction. $\square$

8.2 Left Denominator Sets

We now want to think about relaxing the constraint that we only try to invert regular elements. Note that if S is a m.c. subset of R such that not all elements are regular and $\phi : R \rightarrow S^{-1}R$ is a localization with $\phi(S) \subset (S^{-1}R)^\times$ then whenever $sa = 0$ with $a \in R$ and $s \in S$, we have $\phi(a) = \phi(s)^{-1}\phi(sa) = 0$. So $\ker \phi \supset t_S(R)$.

This motivates the following definition.

Definition 8.7. *Let R be a ring and $S \subset R$ a m.c. subset. A **left ring of fractions** for R with respect to S is a ring T equipped with a ring homomorphism $\phi : R \rightarrow T$ such that*

1. $\phi(S) \subset T^\times$
2. $\ker(\phi) = t_S(R)$
3. every element of T is of the form $\phi^{-1}(s)\phi(a)$ with $s \in S$ and $a \in R$

Note that in the former definition we require S consisting of regular elements, so $t_S(R) = 0$ and ϕ injective is automatic.

Definition 8.8. We say a m.c. subset S of R is **left reversible** if for every $a \in R$ such that $as = 0$ for some $s \in S$ there is $t \in S$ such that $ta = 0$, i.e. $a \in t_S(R)$.

Note that for s regular $as = 0 \implies a = 0$. So this is also automatic in our former settings.

Lemma 8.5. Let S be a m.c. subset of R and suppose that a left ring of fractions $\phi : R \rightarrow T$ exists with respect to S . Then S is a left Ore set and is left reversible. And $\phi(t)^{-1}\phi(a) = \phi(s)^{-1}\phi(b)$ if and only if there is $c, d \in S$ such that $ca = db, ct = ds$

This is similar to lemma 8.2 but we need to use $\ker(\phi) = t_S(R)$. To show S is left reversible, suppose $as = 0$ for $a \in R, s \in S$. Then $\phi(as) = \phi(a)\phi(s) = 0 \implies \phi(a) = 0$ since $\phi(s)$ is a unit. Then $a \in t_S(R)$, and so S is reversible.

Definition 8.9. We call a left reversible left Ore set S a **left denominator set**

Proposition 8.7. If R has the acc. on left annihilators of single elements then every Ore set S is a left denominator set.

Proof. Suppose $as = 0$ with $a \in R$ and $s \in S$. Consider the ascending chain $\text{lann}_R(s) \leq \text{lann}_R(s^2) \leq \dots$. By assumption $\text{lann}_R(s^n) = \text{lann}_R(s^{n+1})$ for some n . Now $bs^n = ta$ for some $t \in S$ and $b \in R$ since S is left Ore. Then $bs^{n+1} = tas = 0 \implies b \in \text{lann}_R(s^{n+1}) = \text{lann}_R(s^n)$. So $ta = bs^n = 0$ and so S is left reversible. $\square$

Theorem 8.2. (Gabriel). Suppose that S is a m.c. subset of a ring R . There exists a left ring of fractions of R with respect to S if and only if S is a left denominator set.

This is reduced to proposition 8.2 by replacing R with $R/t_S(R)$ since $R/t_S(R)$ is S -torsionfree and S again consists of regular elements in $R/t_S(R)$.

So for S not consisting of regular elements, the necessary and sufficient condition for the existence of the left ring of fractions w.r.t. S is S is a left denominator set, i.e. left Ore and left reversible. As before we will denote such a left ring of fractions $S^{-1}R$

Lemma 8.6. For M, N two $S^{-1}R$ -modules, every R -homomorphism $f : M \rightarrow N$ is also a $S^{-1}R$ -homomorphism

Proof. It suffices to show that $f(s^{-1}m) = s^{-1}f(m)$ for all $s \in S, m \in M$. Now $s(f(s^{-1}m) - s^{-1}f(m)) = f(s \cdot s^{-1}m) - f(m) = 0$. So $f(s^{-1}m) - s^{-1}f(m) \in N$ is S -torsion. But N is S -torsionfree since N is $S^{-1}R$ -module. So $f(s^{-1}m) = s^{-1}f(m)$ as required. $\square$

We'd now like consider localisation of modules with respect to a left denominator set.

Definition 8.10. Let S be a left denominator set in a ring R and M a left R -module. A **module of fractions** of M with respect to S consists of a left $S^{-1}R$ -module $S^{-1}M$ together with an R -module homomorphism $\iota_M : M \rightarrow S^{-1}M$ such that

1. each element of $S^{-1}M$ is of the form $\phi(s)^{-1}\iota_M(m)$ for some $s \in S$ and $m \in M$
2. $\ker(\iota_M) = t_S(M)$

Proposition 8.8. (Universal property for modules of fractions) Suppose R is a ring and S is a left denominator set in R . Let M be a left R -module and $\iota_M : M \rightarrow S^{-1}M$ makes $S^{-1}M$ into a module of fractions of M with respect to S . If N is any left $S^{-1}R$ -module then pre-composition with ι_M induces a bijection

$$\text{Hom}_{S^{-1}R}(S^{-1}M, N) \xrightarrow{\sim} \text{Hom}_R(M, N)$$

The injectivity is clear. To see surjectivity, take $\beta \in \text{Hom}_R(M, N)$. Since N is a $S^{-1}R$ -module, N is S -torsionfree and so $t_S(M) \subset \ker(\beta)$. So β induces $\bar{\beta} : M/t_S(M) \rightarrow N$. We know $M/t_S(M) \rightarrow S^{-1}M$ induced by ι_M is a monomorphism. Hence if we take $E(N)$ the injective hull of N , then $\bar{\beta}$ extends to $\gamma : S^{-1}M \rightarrow E(N) \in \text{Hom}_R(S^{-1}M, E(N))$. We need to show that γ lands in N .

Recall that $N \leq_e E(N)$ and N torsionfree implies $E(N)$ torsionfree for S left Ore set. For $s \in S$ and $m \in M$, $s \cdot \gamma(\phi^{-1}(s)\iota_M(m)) = \gamma(\iota_M(m)) = \beta(m)$. Since N is a $S^{-1}M$ -module, let $\phi^{-1}(s)\beta(m) = n \in N$. Then $s \cdot n = \phi(s)n = \beta(m)$. So $s \cdot (n - \gamma(\phi^{-1}(s)\iota_M(m))) = 0$ and hence $n = \gamma(\phi^{-1}(s)\iota_M(m))$ since $E(N)$ S -torsionfree. So γ lands in N .

Remark 8.5. As usual it follows that, if it exists, a module of fractions of M with respect to S is unique up to unique isomorphism.

Theorem 8.3. Let R be a ring, S a left denominator set and M a left R -module. Then there exists a module of fractions $S^{-1}M$ for M .

We will omit this proof. For the rest of this section S is a left denominator set in a ring R .

Definition 8.11. We say that a left R -module M is **S -divisible** if $M \rightarrow M, m \mapsto sm$ is injective for all $s \in S$.

Remark 8.6. It is clear that if M is an S -torsionfree S -divisible R -module then $\iota_M : M \rightarrow S^{-1}M$ is an isomorphism of R -modules.

Proposition 8.9. The universal property for a module of fractions enables us to define an exact functor from left R -modules to left $S^{-1}R$ -modules that sends M to $S^{-1}M$

The varification of exactness is as in the commutative case.

Remark 8.7. One can show that $S^{-1}M \cong S^{-1}R \otimes_R M$. So this proposition can be viewed as the statement that $S^{-1}R$ is a flat right R -module. This says that localisation in the sense we have been discussing is as well-behaved as in the commutative case.

Corollary 8.5. Every injective $S^{-1}R$ -module is injective as an R -module.

Proof. Suppose E is an injective $S^{-1}R$ module. Let $\alpha \in \text{Hom}_R(M, N)$ a monomorphism and $\phi \in \text{Hom}_R(M, E)$. By proposition 8.9, $S^{-1}\alpha \in \text{Hom}_{S^{-1}R}(S^{-1}M, S^{-1}N)$ (which the unique map satisfying $S^{-1}\alpha \circ \iota_M = \iota_N \circ \alpha$) is a monomorphism. Also by proposition 8.8, there is $\varphi \in \text{Hom}_{S^{-1}R}(S^{-1}M, E)$ such that $\varphi \circ \iota_M = \phi$. Then by inctivity of E as a $S^{-1}R$ -module, φ extends to $\theta \in \text{Hom}_{S^{-1}R}(S^{-1}N, E)$ such that $\theta \circ S^{-1}\alpha = \varphi$. Hence

$$\theta \circ \iota_N \circ \alpha = \theta \circ S^{-1}\alpha \circ \iota_M = \varphi \circ \iota_M = \phi$$

So $\theta \circ \iota_N$ extends ϕ , and E is an injective R -module. □

Corollary 8.6. *For every S -torsionfree injective R -module E , the map $\iota_E : E \rightarrow S^{-1}E$ is an R -module isomorphism and $S^{-1}E$ is an injective $S^{-1}R$ -module.*

Proof. It suffices to show that E is S -divisible. Suppose $s \in S$ and $x \in E$. Then since S is left-reversible and E is S -torsionfree, $\text{lann}_R(s) \leq t_S(R) \leq \text{lann}_R(x)$. Thus we define $\alpha : Rs \rightarrow E$ given by $rs \mapsto rx$. By injectivity of E this extends to $\alpha' : R \rightarrow E$ with $s\alpha'(1) = \alpha(s) = x$. So E is S -divisible.

Now if $N \leq M$ two $S^{-1}R$ -modules, then by identifying E and $S^{-1}E$ we can view an element of $\text{Hom}_{S^{-1}R}(N, S^{-1}E)$ as an element of $\text{Hom}_R(N, E)$. By injectivity of E , it extends to an element of $\text{Hom}_R(M, E) = \text{Hom}_R(M, S^{-1}E)$. By an exercise we know the R -linear maps between $S^{-1}R$ -modules are indeed $S^{-1}R$ linear. $\square$

Suppose R is a ring and S is any m.c. subset of R . We define $R\langle i_s, s \in S \rangle / (i_s \cdot s - 1, s \cdot i_s - 1, s \in S)$. This is the formal construction of $S^{-1}R$. We can check that it has the universal property for the ring of fractions, so $S^{-1}R$ as defined here is canonically isomorphic to our previous definition. However localisation in this generality is not well-behaved.

8.3 Exercise

Proposition 8.10. *Show that if R is a left noetherian domain then R is uniform as a left R -module. Deduce directly that $R - \{0\}$ is a left Ore set*

Proof. Suppose $I, J \triangleleft_l R$ two nonzero left ideals with $I \cap J = 0$. Now given any $0 \neq K \triangleleft_l R$, take $0 \neq k \in K$. Then $Ik, Jk \triangleleft_l R$ are two nonzero left-ideals contained in K since I, J, k non zero and R is domain. And $Ik \cap Jk = (I \cap J)k = 0$. So we just proved to following claim: for any $0 \neq K \triangleleft_l R$, there exist two non-zero left ideals $0 \neq K_1, K_2 \subset K$ such that $K_1 \cap K_2 = 0$, and hence $K_1 \oplus K_2 \subset K$. Repeat this process and we can get an infinite direct sum of non-zero left ideals in R , which contradicts the noetherianity of R .

Now given $a \in R, s \in R - \{0\}$. If $a = 0$ then $sa = 0 = 0s$. So assume $a \neq 0$. Then by the first part we have $Ra \cap Rs \neq 0$, i.e. there are $r_1, r_2 \neq 0$ such that $r_1a = r_2s$. $\square$

Proposition 8.11. *For $R = A_1(k)$ the Weyl algebra, show that $\{1, x, x^2, \dots\}, \{1, y, y^2, \dots\}, k[x] \setminus 0, k[y] \setminus 0$ are left Ore sets in R .*

Proof. We can use lemma 1.2 to give a quick proof of showing $\{1, x, x^2, \dots\}$ is left Ore. We claim that ad_x is **locally nilpotent**, i.e. for any $r \in R$, $\text{ad}_x^i(r) = 0$ for some i . Note that $\text{ad}_x(xr) = x\text{ad}_x(r)$ and $\text{ad}_x(rx) = \text{ad}_x(r)x$, so it is reduced to when r is a power of y . Now $\text{ad}_x(y^n) = -ny^{n-1}$ and $\text{ad}_x(y) = -1, \text{ad}_x^2(y) = 0$, so this follows immediately from induction. Also ad_x is linear, so it suffices to take N maximal among those n on which ad_x^n is 0 on one monomial of r . So the claim follows. And we only need to take $n = N + 1$ in lemma 1.2 to get $x^n r = bx$ for some $b \in R$. $\square$

Proposition 8.12. *Suppose that α is ring endomorphism of R .*

1. *Show that there is a ring $S = R[y; \alpha]$ that can be equipped with the data of a injective homomorphism $\phi : R \rightarrow S$ and an element $y \in S$, such that for every ring T there is a bijection between the set of ring homomorphisms $\varphi : S \rightarrow T$ and the set of pairs (θ, t) such that $\theta : R \rightarrow T$ is a ring homomorphism and $t \in T$ satisfying $t\theta(r) = \theta(\alpha(r))t$ for all r . This bijection is given by $\varphi \mapsto (\varphi\phi, \varphi(y))$*
2. *Show that if R is left noetherian and α is an automorphism then $R[y; \alpha]$ is left noetherian*

3. Show that if $R = k(x)$ is the field of rational functions in one variable over k and α is defined by $\alpha(f(x)) = f(x^2)$ then $R[y; \alpha]$ is a left noetherian domain but is not right noetherian. Show moreover that the non-zero elements of $R[y; \alpha]$ form a left Ore set but not a right Ore set.

Another very confusing description of some universal property

Proof. We only give the construction of S . Consider $E = \text{End}_{\mathbb{Z}}(R^{\mathbb{N}})$. Consider $\phi : R \rightarrow E; r \mapsto r \cdot (-)$, and $y \in E$ given by $y(r_1, r_2, \dots) = (0, \alpha(y_1), \alpha(y_2), \dots)$. Let $S = \langle \phi(R), y \rangle \subset E$ a subring. We claim that every element of S is uniquely of the form $\sum_{i=0}^n \phi(r_i) y^i$ for some $r_i \in R$. To see this, it is enough to show that $y\phi(r)$ has this form for all r . But $y\phi(r)(r_1, r_2, \dots) = y(rr_1, rr_2, \dots) = (0, \alpha(rr_1), \dots) = \phi(\alpha(r))y(r_1, \dots)$. So $y\phi(r) = \phi(\alpha(r))y$. Using this we see the map between the set of ring homomorphisms $\varphi : S \rightarrow T$ and the set of pairs (θ, t) such that $\theta : R \rightarrow T$ is a ring homomorphism and $t \in T$ satisfying $t\theta(r) = \theta(\alpha(r))t$ for all r given $\varphi \mapsto (\varphi\phi, \varphi(y))$ is well-defined. The inverse is defined by $\varphi(\sum \phi(r_i) y^i) = \sum \theta(r_i) t^i$.

For (2) to apply Hilbert's basis theorem need to check $R + Ry = R + yR$ by identifying R and $\phi(R)$. This clear since $y\phi(r) = \phi(\alpha(r))y$ and α is automorphism. $\square$

Proposition 8.13. Show that if $V = k^{\mathbb{N}}$ is a k -vector space with a countable basis then $S = \{\alpha \in \text{End}_k(V) \mid \alpha \text{ is injective}\}$ is a multiplicatively closed subset of $\text{End}_k(V)$ all of whose elements have left inverses. Show that S is a left Ore set that is not left reversible

Proof. α has left inverse because $\alpha(e_i)$ are linear independent and hence can be extended to a basis of V , and so we can define its left inverse by the basis directly. Now for $\alpha \in S$ and $\beta \in \text{End}_k(V)$, take $\delta = \beta\alpha^{-1}$ and $\gamma = \text{id}_V \in S$ and we have $\delta\alpha = \gamma\beta$. Now take β given by $e_1 \mapsto e_1, e_i \mapsto 0$ for $i \neq 1$ and $\alpha \in S$ shifting by 1. Then $\beta\alpha = 0$. But there doesn't exist $\alpha' \in S$ such that $\alpha'\beta = 0$ since α' has a left inverse $\implies \beta = 0$. $\square$

Proposition 8.14. Suppose that R is a semiprime left Goldie ring, S is the set of its regular elements and M is a left R -module.

1. Using Goldie's Regular Element Lemma show that if M is S -torsionfree and S -divisible then M is injective. Deduce that the classical left quotient ring Q of R is an injective hull of R
2. Show that if M is not S -torsion then M has a uniform submodule isomorphic to a left ideal of R
3. Give an example to show that it is not always the case that all S -torsionfree uniform modules

are isomorphic to left ideals of R .

Proof. (1) By Baer's criterion, it is enough to show that for any $I \triangleleft_l R$ and R -module homomorphism $\phi : I \rightarrow M$, it can extend to $R \rightarrow M$.

First we consider when $I \leq_e R$. By Goldie's Regular Element Lemma, I contains a regular element s . Now M is S -divisible gives $m \in M$ such that $sm = \phi(s)$. Then we define $\varphi : R \rightarrow M$ given by $1 \mapsto m$. To see that φ extends ϕ , it suffices to show that $\varphi(r) = rm = \phi(r)$ for any $r \in I$. Note that Goldie's theorem gives that S is left Ore. So there is $s' \in S$ and $r' \in R$ such that $s'r = r's \in I$. So $\phi(r's) = r'\phi(s) = r'sm = \phi(s'r) = s'\phi(r)$. But $s'r = r's \implies s'rm = r'sm$. So $s'rm = s'\phi(r) \implies rm = \phi(r)$ since M is S -torsion free.

In general, we take J maximal subject to $I \cap J$. Then $I \oplus J \leq_e R$. For any $\phi : I \rightarrow R$, we can extend it to $I \oplus J$ by mapping the whole J to 0. So we are reduced to the first case.

For the classical quotient ring Q of R , Q is an injective R -module by the first part because Q is S -divisible and S -torsionfree. Take any $0 \neq q \in Q$, then there is $s \in S$ such that $0 \neq sq \in R$. Hence $0 \neq Rq \cap R$ and $R \leq_e Q$

(2) M is not torsion implies that there is $m \in M$ such that $\text{lann}_R(m) \cap S = 0$. By Goldie's regular element lemma, $\text{lann}_R(m)$ is not essential in R . Take $0 \neq J \triangleleft_l R$ maximal subject to $\text{lann}_R(m) \cap J = 0$. Then Jm is a submodule of M and $J \rightarrow Jm; r \mapsto rm$ is an isomorphism. Now we can take a uniform submodule $U \cong Um$ of J using $\text{rank}(R) < \infty$

(3) Take $R = \mathbb{Z}$ and $\mathbb{Q}$ is a torsionfree uniform R -module. It is not finitely generated and hence is not isomorphic to any left ideal of R which is principal. $\square$

Proposition 8.15. *Show that if k is a field and R is the ring of endomorphism of a vector space V with a countably infinite basis, and S is the set of injective maps in R then $S^{-1}R = 0$.*

Proof. We can think of $V = k[t]$. Note that $k[t] = k[t^2] \oplus tk[t^2]$ and so $\text{id}_{k[t]} = \pi_1 + \pi_2$ where π_i 's are the projections w.r.t this direct sum. Then there are two injections $s_1 : k[t] \rightarrow k[t^2]; f(t) \mapsto f(t^2)$ and $s_2 : k[t] \rightarrow tk[t^2]; f(t) \mapsto tf(t^2)$. Then $\pi_1 \circ s_2 = \pi_2 \circ s_1 = 0$. But $s_1, s_2 \in (S^{-1}R)^\times$, so $\pi_1 = \pi_1 \circ s_2 \circ s_2^{-1} = 0 \in S^{-1}R$ and similarly $\pi_2 = 0 \in S^{-1}R$. So $\text{id}_{k[t]} = \pi_1 + \pi_2 = 0 \in S^{-1}R$. $\square$

Proposition 8.16. 1. *For $r \in R$, we say ad_r is **locally nilpotent** if for all $a \in R$ there is $n \geq 0$ such that $\text{ad}_r^n(a) = 0$. Suppose that S is a multiplicatively closed subset of R . Show that if for every $s \in S$, ad_s is locally nilpotent then S is a left denominator set in R .*

2. *Suppose now that there is $t \in Z(R)$ such that $t^n = 0$ and R/tR is commutative. Deduce that every multiplicatively closed subset $S \in R$ is a left denominator set.*

3. *Show that if $S \subset R$ is multiplicatively closed and $\bar{S} = S + tR \subset R/tR$ then there is a natural bijection $\text{Spec}(\bar{S}^{-1}R/tR) \rightarrow \text{Spec}(S^{-1}R)$*

Proof. (1) Let $a \in R, s \in S$. Suppose $\text{ad}_s^n(a) = 0$. By lemma 1.2, $s^n a = bs$ for some $b \in R$. So S is a left Ore set. If $as = 0$, then $\text{ad}_s(a) = sa$. By induction we see that $\text{ad}_s^i(a) = s^i a$ for all i . Hence $\text{ad}_s^n(a) = s^n a = 0$ and so S is left reversible

(2) By (1) it suffices to show that ad_s is locally nilpotent for all $s \in S$. Take $a \in R$. R/tR is commutative implies that $\text{ad}_s(a) \in tR$. Say $\text{ad}_s(a) = ta_1$. Then $\text{ad}_s^2(a) = \text{ad}_s(ta_1) = t \cdot \text{ad}_s(a_1)$ since $t \in Z(R)$. So $\text{ad}_s^2(a) \in t^2R$. By induction we see that $\text{ad}_s^n(a) \in t^n R = 0$ by assumption.

(3) It suffices to show that $\bar{S}^{-1}R/tR \cong S^{-1}R/tS^{-1}R$. Granting this, since $t \in Z(R)$ we have $(tS^{-1}R)^n = t^n S^{-1}R = 0$. So $\text{Spec}(S^{-1}R/tS^{-1}R) \cong \text{Spec}(S^{-1}R)$. Now we need to show $\phi : R/tR \rightarrow S^{-1}R/tS^{-1}R$ given by $r+tR \mapsto r+tS^{-1}R$ satisfies the requirement in definition 8.7 so that $S^{-1}R/tS^{-1}R$ is the left ring of fractions for R w.r.t S and by universal property the isomorphism follows. That is, to check $\phi(\bar{S}) \in (S^{-1}R/tS^{-1}R)^\times$, $\ker \phi = t_{\bar{S}}(R/tR)$ and every element of $S^{-1}R/tS^{-1}R$ has the form of $\phi^{-1}(\bar{s})\phi(\bar{a})$, which is not so fun. $\square$

9 Filtrations and Associated Graded Rings

Definition 9.1. *Let R be a ring. A $(\mathbb{Z})$ -filtration on R is a set of additive subgroups $(R_i)_{i \in \mathbb{Z}}$ such that*

1. $R_i \subset R_{i+1}$ for all $i \in \mathbb{Z}$
2. $R_i \cdot R_j \subset R_{i+j}$
3. $1 \in R_0$
4. $\cup_{i \in \mathbb{Z}} R_i = R$

If R has a filtration, we say that R is a **filtered ring**. The filtration on R is **positive** if $R_i = 0$ for all $i < 0$.

Note that the axioms imply that R_0 is a subring of R and that each R_i is a left and right R_0 -module. Note also that $\cap_{i \in \mathbb{Z}} R_i$ is always an ideal in R .

Example 14. Suppose R is a finitely generated k -algebra with generating set $\{x_1, \dots, x_n\}$. Define $R_0 = k$ and let R_i be the k -subspace spanned by words with length at most i . Also define $R_i = 0$ for $i < 0$.

Definition 9.2. A $(\mathbb{Z})$ -**graded ring** is a ring S which can be written as $S = \oplus_{i \in \mathbb{Z}} S_i$ for some additive subgroups S_i , satisfying $S_i \cdot S_j \subset S_{i+j}$ and $1 \in S_0$. S_i is called the **i -th homogeneous component** of S and $s \in S$ is called **homogeneous** if $s \in S_n$ for some n .

Definition 9.3. Let R be a filtered ring with filtration $(R_i)_{i \in \mathbb{Z}}$. Define

$$gr(R) = \oplus_{i \in \mathbb{Z}} R_i / R_{i-1}$$

Equip grR with multiplication, which is given on homogeneous components by $(r + R_{i-1}, s + R_{j-1}) \mapsto rs + R_{i+j-1}$ and on the whole of grR by bilinear extension. Then grR becomes a graded ring called the **associated graded ring** of R .

One should think of grR as an approximation to the ring R which is often easier to understand but nonetheless contains useful information about the ring R itself, including noetherianity.

Proposition 9.1. Let $\mathfrak{g}$ be a k -Lie algebra over a commutative ring k . Equip $U(\mathfrak{g})$ with the positive filtration

$$U(\mathfrak{g})_0 = k, U(\mathfrak{g})_1 = k + \mathfrak{g}, U(\mathfrak{g})_n = (U(\mathfrak{g})_1)^n \text{ for } n \geq 1$$

Then there is a surjective homomorphism of k -algebras $\phi : Sym(\mathfrak{g}) \rightarrow grU(\mathfrak{g})$ given by $\phi(x) = x + U(\mathfrak{g})_0$ for $x \in \mathfrak{g}$

Remark 9.1. We call this the *PBW filtration*. The PBW theorem says that whenever $\mathfrak{g}$ is a free k -module (or even projective, i.e. a direct summand of a free module) then ϕ is an isomorphism. In particular, this is an isomorphism when k is a field.

Proof. By the universal property of Sym , to show ϕ defines a ring homomorphism it suffices to show the $\phi(x)\phi(y) = \phi(y)\phi(x)$ for $\phi : \mathfrak{g} \rightarrow gr(U(\mathfrak{g}))$, $\phi(x) = x + U(\mathfrak{g})_0$. But in $U(\mathfrak{g})$ $xy - yx = [xy] \in U(\mathfrak{g})_1$. So $\phi(x)\phi(y) - \phi(y)\phi(x) = (x + U(\mathfrak{g})_0)(y + U(\mathfrak{g})_0) - (y + U(\mathfrak{g})_0)(x + U(\mathfrak{g})_0) = 0 \in U(\mathfrak{g})_2/U(\mathfrak{g})_1$. So it factors through $Sym(\mathfrak{g})$.

To show that ϕ is surjective, it's sufficient to show that $a + U(\mathfrak{g})_{n-1}$ lies in $im\phi$ for any $a \in U(\mathfrak{g})_n = (U(\mathfrak{g})_1)^n$. But this is clear since for $x_1, \dots, x_n \in \mathfrak{g}$ then $\phi(x_1 \dots x_n) = x_1 \dots x_n + U(\mathfrak{g})_{n-1}$ $\square$

Proposition 9.2. Let $R = A_n(k)$. Set $R_0 = k[x_1, \dots, x_n]$ and R_i the R_0 -module generated by monomials in y_k 's with degree at most i . Then $(R_i)_{i \in \mathbb{Z}}$ is a positive filtration on R and $grR \cong k[X_1, \dots, X_n, Y_1, \dots, Y_n]$

We observe that $y^\beta R_0 \subset R_{|\beta|}$ with $\beta \in \mathbb{N}_0^n$. The relations $y_i x_j - x_j y_i = \delta_{ij}$ become trivial in $gr(R)$, since $1 \in R_0$. So $gr(R)$ is commutative.

Proposition 9.3. Suppose R is a positively filtered ring such that grR is left noetherian. Then R is left noetherian.

For I an ideal in R , we can define grI the associated homogeneous ideal of I . Then grI is generated by a finite set of homogeneous element $\{X_1, \dots, X_n\}$. Take $x_i \in I \cap R_{n_i} \setminus R_{n_i-1}$ such that $x_i + R_{n_i-1} = X_i$. We can show that they generate I , by showing that $I \cap R_n \subset \sum R x_i$ by induction.

Remark 9.2. $A_n(k)$ is noetherian combining proposition 9.2 and 9.3.

Corollary 9.1. $U(\mathfrak{g})$ is noetherian whenever $\dim_k(\mathfrak{g}) < \infty$

By proposition 9.1 $grU(\mathfrak{g})$ is a quotient of $Sym(\mathfrak{g})$ which is a commutative finitely generated k -algebra and hence noetherian. So $U(\mathfrak{g})$ is noetherian.

Definition 9.4. Let R be a left noetherian ring. An (exact) **dimension function** for R is a rule which assigns to every finitely generated R -module M a number $d(M) \in \mathbb{N}_0 \cup -\infty$ such that $d(M) = \max\{d(L), d(N)\}$ for every short exact sequence $0 \rightarrow L \rightarrow M \rightarrow N \rightarrow 0$, and $d(M) = -\infty$ if and only if $M = 0$.

Now recall that the **Krull dimension** of a commutative ring A is the supremum of the lengths of chains of prime ideals in A . Let M be a finitely generated A -module. Then we define $Kdim(M) = Kdim(A/ann_A(M))$ the Krull dimension of M

In general, $Kdim(A)$ can be infinite, even if A is noetherian. However it is always finite if A is a finitely generated comcommutative algebra over a field k . Indeed it is bounded by the number of generators.

Remark 9.3. The Krull dimension is a dimension function for A . For a finitely generated A -module M , we define its **support** is $Supp(M) = \{p \in Spec A \mid M_p \neq 0\} = V(Ann_R(M))$. So $Kdim(M)$ is the supremum of the lengths of chains of prime ideals in $Supp(M)$. Also by exactness of localization, we have $Supp(M) = Supp(L) \cup Supp(N)$ for a s.e.s. With this we can see Krull dimension is a dimension function.

Now, let's return to the non-commutative setting and seek a well-behaved dimension function in the case where the ring in question doesn't necessarily have many two-sided ideals.

Definition 9.5. Let R be a filtered ring with filtration $(R_n)_{n \in \mathbb{Z}}$ and let M be a left R -module. A **filtration** on M is a set $(M_n)_{n \in \mathbb{Z}}$ of additive subgroups of M satisfying

1. $M_i \subset M_{i+1}$ for all i
2. $R_i \cdot M_j \subset M_{i+j}$
3. $M = \cup_{n \in \mathbb{Z}} M_n$

Example 15. Let M be a left R -module with generating set X . Then $M_n = R_n \cdot X$ for all $n \in \mathbb{Z}$ gives a filtration of M , known as a **standard filtration** with respect to X .

We can similarly define **graded left S-module** and **graded left ideal** of S for a graded ring S .

For a filtered ring R and a filtered R -module M , we can define its **associated graded module** $grM = \bigoplus_{n \in \mathbb{Z}} M_n / M_{n-1}$, which is naturally a grR -module.

Our next goal will be to define a well-behaved dimension function for certain filtered non-commutative rings via $d(M) = Kdim(grM)$ but we'll need to establish this doesn't depend on the filtration of M or more precisely to restrict the class of filtrations we consider. For this, we first need to make a digression to study Rees rings and good filtrations.

Definition 9.6. Given a filtered ring R with filtration $(R_n)_{n \in \mathbb{Z}}$ and a filtered R -module M with filtration $(M_n)_{n \in \mathbb{Z}}$, the **Rees ring** is the following graded subring $\tilde{R}$ of the ring of Laurent polynomials $R[t, t^{-1}]$:

$$\tilde{R} = \bigoplus_{n \in \mathbb{Z}} R_n t^n \subset \bigoplus R t^n = R[t, t^{-1}]$$

The **Rees module** $\tilde{M}$ of M is the abelian group $\tilde{M} = \bigoplus_{n \in \mathbb{Z}} M_n t^n$ which is a $\tilde{R}$ -module with the graded action $R_n t^n \times M_m t^m \rightarrow M_{n+m} t^{n+m}$, $(rt^n, at^m) \mapsto rat^{n+m}$

Rees rings and Rees modules interpolate between R and grR respectively M and grM .

Lemma 9.1. Let R and M be as above. Then

1. $\tilde{R}/t\tilde{R} \cong grR$ as graded rings
2. $\tilde{R}/(t-1)\tilde{R} \cong R$ as rings
3. $\tilde{M}/t\tilde{M} \cong grM$ as graded left $\tilde{R}$ -modules
4. $\tilde{M}/(t-1)\tilde{M} \cong M$ as $\tilde{R}$ -modules

Proof. (1) $\tilde{R}/t\tilde{R} = \bigoplus R_n t^n / \bigoplus R_{n-1} t^n \cong \bigoplus (R_n / R_{n-1}) t^n \cong \bigoplus R_n / R_{n-1} = grR$

(2) Consider the map $\pi : \tilde{R} \rightarrow R$ given by $\pi(\sum a_n t^n) = \sum a_n$. It is clear that π is surjective. WTS that $\ker \pi = (t-1)\tilde{R}$.

Certainly $\pi(t-1) = 0$. Now suppose $a = \sum a_n t^n \in \ker \pi$. Then $\sum a_n = 0$ and so $a = a - \sum a_n = \sum a_n(t^n - 1)$. Since $t^n - 1$ is a multiple of $t - 1$, we have $a = b(t-1)$ for some $b \in R[t, t^{-1}]$. Let $b = \sum b_n t^n$. Then $\sum a_n t^n = \sum b_n t^n(t-1) = \sum (b_{n-1} - b_n) t^n$. Then $(b_{n-1} - b_n) \in R_n$. We prove by induction that $b_n \in R_n$. Note that $b_n = 0$ for $n < 0$. Suppose $b_n \notin R_n$ for n least. Then $b_{n-1} \in R_{n-1} \subset R_n$ and hence $b_n = b_{n-1} - a_n \in R_n$, contradiction. So $b \in \tilde{R}$ and hence $\ker \pi = (t-1)\tilde{R}$ $\square$

Definition 9.7. Let R be a filtered ring and let M be a left R -module. A filtration $(M_n)_{n \in \mathbb{Z}}$ is a **good filtration** if the associated Rees module $\tilde{M}$ is finitely generated over $\tilde{R}$.

Proposition 9.4. A filtration is good if and only if there are $k_1, \dots, k_n \in \mathbb{Z}$ and $x_i \in M_{k_i}$ such that $M_n = \sum_{i=1}^k R_{n-k_i} x_i$ for all n . In particular, **M and grM are finitely generated as R and grR -modules.**

Proof. If $\tilde{M}$ is finitely generated, it admits a homogeneous generating set $\{x_1 t^{k_1}, \dots, x_n t^{k_n}\}$ with $x_i \in M_{k_i}$. Then the n -th homogeneous component of $\tilde{M}$ is $M_n t^n = \sum R_{n-k_i} t^{n-k_i} x_i t^{k_i} \sum R_{n-k_i} x_i t^n$. So $M_n = \sum_{i=1}^k R_{n-k_i} x_i$.

Conversely, if $M_n = \sum_{i=1}^k R_{n-k_i} x_i$ for any n , then $\{x_i t^{k_i}, i = 1, \dots, n\}$ is a generating set for $\tilde{M}$ so the filtration is good. $\square$

Corollary 9.2. Every finitely generated module over a filtered ring has a good filtration.

Let X be a finite generating set for M . Then the standard filtration of M with respect to X is good since $M_n = R_n \cdot X$. In this case the generators in X are of degree 0.

Definition 9.8. Two filtrations are **algebraically equivalent** if there exist $c, d \in \mathbb{Z}$ such that $M_n \subset M'_{n+c}$ and $M'_m \subset M_{m+d}$ for all $m, n \in \mathbb{Z}$

Lemma 9.2. All good filtrations are equivalent.

Indeed, we have $M_n = \sum R_{n-k_i} x_i$ for all n . Then there exists c such that $x_i \in M'_{k_i+c}$. Hence $M_n \subset M'_{n+c}$ for all n . By symmetry, there is d such that $M'_m \subset M_{m+d}$ for all m .

9.1 Characteristic Variety

Theorem 9.1. *Let R be a filtered ring such that grR is commutative and noetherian, and let M be a finitely generated R -module. Suppose that (M_n) and (M'_n) are two good filtrations on M , and let grM , $gr'M$ be the respective associated graded modules. Then for $P \in Spec(grR)$, $Ann_{grR} grM \subset P$ if and only if $Ann_{grR} gr'M \subset P$, or equivalently, their radicals coincide.*

Proof. By lemma 9.2 (M_n) and (M'_n) are equivalent. Taking the maximal of c and d , we may assume that $M_n \subset M'_{n+c}$ for all n and $M'_m \subset M_{m+c}$ for all m .

Let $I = Ann(grM)$ and $J = Ann(gr'M)$. WTS $(I + J)/J \triangleleft grR/J$ is nilpotent. Since grR is commutative and noetherian, $(I + J)/J$ is finitely generated and hence it suffices to show that every element of $(I + J)/J$ is nilpotent. Since I and J are both graded ideals it even suffices to show that all homogeneous elements are nilpotent.

Suppose $a + R_{k-1} \in I \cap (R_k/R_{k-1})$. Then $aM_n \subset M_{n+k-1}$ for all n since $a + R_{k-1} \in Ann(grM)$. So $a^{2c+1}M'_m \subset a^{2c+1}M_{m+c} \subset M_{m+(2c+1)(k-1)+c} \subset M'_{m+(2c+1)k-1}$ for all m . Hence $(a + R_{k-1})^{2c+1} \in J$ as required.

Now if $J \subset P$ for some $P \in Spec(grR)$, we know that $I^N \subset (I + J)^N \subset J \subset P$ for some N and hence $I \subset P$. The reverse implication will then follow by symmetry. $\square$

Definition 9.9. *Let R be a filtered ring such that grR is a finitely generated commutative algebra over a field k , and let M be a finitely generated R -module. Choose a good filtration on M . The **characteristic variety** of M is $Ch(M) = \{P \in Spec(grR) \mid Ann(grM) \subset P\} = V(Ann(grM))$. The **dimension** of M is $Kdim(grM)$*

By theorem 9.1, $Ch(M)$ and $dim(M)$ are independent of the choice of good filtrations on M .

Now we show that d really is dimension function

Definition 9.10. *Let N be a submodule of a filtered left R -module M . The **subspace filtration** is $(N_n = N \cap M_n)$ on N . The **quotient filtration** is $((M/N)_n = (M_n + N)/N)$ on M/N*

Remark 9.4. There is a natural s.e.s. $0 \rightarrow grN \rightarrow grM \rightarrow gr(M/N) \rightarrow 0$ regarding these filtrations.

Proposition 9.5. *Let R be a filtered ring such that grR is a finitely generated commutative algebra over a field k and $\tilde{R}$ is left noetherian. Then $M \mapsto d(M) = Kdim(grM)$ is a dimension function for R .*

Proof. Equip M with a good filtration and endow N and M/N with the subspace and quotient filtrations, respectively. Since each level of these filtrations is exact, take direct sum and we have $0 \rightarrow \tilde{N} \rightarrow \tilde{M} \rightarrow \tilde{M}/N \rightarrow 0$. Since the filtration is good, $\tilde{M}$ is finitely generated $\tilde{R}$ -module. As $\tilde{M}$ is left noetherian, $\tilde{N}$ and $\tilde{M}/N$ are finitely generated. So the subspace and quotient filtrations are good. And hence grM , grN , grM/N are all finitely generated. So $0 \rightarrow grN \rightarrow grM \rightarrow gr(M/N) \rightarrow 0$ is an exact sequence of finitely generated modules. Since $d(M) = Kdim(grM)$, d is dimension function since $Kdim$ is. $\square$

Remark 9.5. Indeed we have $Ch(M) = Ch(N) \cup Ch(M/N)$ as the support does.

Remark 9.6. Although $d(M)$ does not depend on the choice of good filtration on M , it may still depend on the choice of filtration on R . However it can be proven, using homological techniques beyond the scope of this course, that in fact it does not.

9.2 Poisson structure

Definition 9.11. Let R be a k -algebra. A **Poisson bracket** on R is a function $\{-, -\} : R \times R \rightarrow R$ such that

1. $\{-, -\}$ makes R a k -Lie algebra, i.e. a Lie bracket
2. $\{x, yz\} = \{x, y\}z + y\{x, z\}$, i.e. $\{x, -\} : R \rightarrow R$ is a derivation on R

We call R equipped with a Poisson bracket a **Poisson algebra**. If R is a graded k -algebra, we say $\{-, -\}$ has degree r if $\{R_n, R_m\} \subset R_{n+m+r}$ for all n, m .

Note that $\{-, y\}$ is also a derivation because Lie bracket is skew-symmetric.

The commutator bracket on every ring is an example of a Poisson bracket. However, it can happen that a commutative ring has an interesting and non-trivial additional Poisson structure

Corollary 9.3. Let R be a filtered ring such that grR is commutative. Then there is a Poisson bracket on grR of degree -1 . Concretely, we define $\{x + R_{n-1}, y + R_{m-1}\} \mapsto [x, y] + R_{n+m-2}$

Proof. Certainly $(x, y) \mapsto [x, y]$ is a Poisson bracket on R . Since grR is commutative, if $x \in R_n, y \in R_m$, $[x, y] + R_{n+m-1} = (x + R_{n-1})(y + R_{m-1}) - (y + R_{m-1})(x + R_{n-1}) = 0$ and hence $[x, y] \in R_{n+m-1}$, so $[-, -]$ descends to $\{-, -\} : R_n/R_{n-1} \times R_m/R_{m-1} \rightarrow R_{n+m-1}/R_{n+m-2}$ given by $\{x + R_{n-1}, y + R_{m-1}\} \mapsto [x, y] + R_{n+m-2}$ and its bilinear extension to $grR \times grR$ is a Poisson bracket of degree -1 because the required equations follow from the corresponding equations for $[-, -]$ and multilinearity. $\square$

Whenever R is a filtered k -algebra so that grR is a commutative k -algebra we will view it as a Poisson k -algebra via $\{x + R_{n-1}, y + R_{m-1}\} \mapsto [x, y] + R_{n+m-2}$

Example 16. Consider $R = A_n(k)$, the n th Weyl algebra filtered by order of differential operator. We've seen that $grR \cong k[X_1, \dots, X_n, Y_1, \dots, Y_n]$ with each X_i in degree zero and Y_j in degree one. We can compute $\{X_i, X_j\} = 0, \{Y_i, Y_j\} = 0, \{Y_i, X_j\} = \delta_{ij}$. And the derivations $\{f, -\} = \sum(\partial f / \partial Y_i) \partial / \partial X_i - \sum(\partial f / \partial X_i) \partial / \partial Y_i$ and so $\{f, g\} = \sum(\partial f / \partial Y_i)(\partial g / \partial X_i) - \sum(\partial f / \partial X_i)(\partial g / \partial Y_i)$

Example 17. Consider $R = U(\mathfrak{g})$ for $\mathfrak{g}$ a k -Lie algebra over a field k with the PBW filtration.

The PBW theorem says that $Sym(\mathfrak{g}) \cong gr(R)$ with $\mathfrak{g} \subset Sym(\mathfrak{g})$ in degree 1 and $\{x, y\} = [x, y]$ for all $x, y \in \mathfrak{g}$. This determines everything.

Definition 9.12. Suppose A a commutative k -algebra and $p : A \rightarrow k$ is a k -algebra homomorphism with kernel m_p . Then **cotangent space** to $Spec(A)$ at p is $T_p^*SpecA = m_p/m_p^2$. The **tangent space** of $SpecA$ at p is $T_pSpecA = Hom_k(m_p/m_p^2, k) \cong Der_k(A, k)$. Here $Der_k(A, k) = \{d : A \rightarrow k \mid d(ab) = p(a)d(b) + d(a)p(b), d \text{ is } k\text{-linear}\}$. Note $d(1) = 0$ so $d \in Der_k(A, k)$ is determined by the values on m and $d(m^2) = 0$

Of course, p is a k -point and this should agree with the definition in algebraic geometry

Also if $\theta : A \rightarrow k[t]/t^2$ is a k -algebra homomorphism with $\theta \equiv p \pmod{t}$ (recall that p here is the k -point). Then $\theta = p + td$ for some k -linear $d : A \rightarrow k$ and hence $\theta(ab) = (p + td)(ab) = p(ab) + t(d(a)p(b) + p(a)d(b))$. So $d \in Der_k(A, k)$

Moreover if A is a Poisson algebra then for each such p the Poisson bracket on A induces an alternating bilinear form $\{-, -\}_p$ on the cotangent space via $\{x + m^2, y + m^2\}_p = p(\{x, y\})$

Example 18. If $R = A_n(k)$ filtered by order of differential operator then for every k -algebra homomorphism $p : grR \rightarrow k$, m_p/m_p^2 has k -basis given by $v_{i,p} = X_i - p(X_i) + m^2$, $w_{i,p} = Y_i - p(Y_i) + m^2$ (it is clear that they are in m). And $\{w_{i,p}, v_{j,p}\}_p = \delta_{ij}$ and 0 elsewhere. (The scalars are in the center, so we can ignore them)

If k is an algebraically closed field and A is a reduced (i.e. semiprime) finitely generated commutative k -algebra then for every $p : A \rightarrow k$, $\dim_k(T_p^* \text{Spec} A) \geq K \dim(A)$. We say p is a **smooth k -point** of $\text{Spec} A$ when the equality holds.

Recall that if R is a filtered ring such that $gr R$ is commutative and M is a finitely generated R -module with a good filtration M_n then for $x + R_{k-1}, y + R_{l-1} \in \text{Ann}(gr M)$, $xM_n \subset M_{n+k-1}, yM_n \subset M_{n+l-1}$ for all n . So $[x, y]M_n = (xy - yx)M_n \subset M_{n+k+l-2}$. Thus $\{x + R_{k-1}, y + R_{l-1}\} \in \text{Ann}(gr(M)) \cap (gr R)_{k+l-1}$

Theorem 9.2. (*Involutivity of the Characteristic Variety, Gabber 1981*) Let R be a filtered ring such that $gr R$ is a commutative noetherian $\mathbb{Q}$ -algebra and let M be a finitely generated R -module. Then

$$\{\sqrt{\text{Ann}(gr M)}, \sqrt{\text{Ann}(gr M)}\} \subset \sqrt{\text{Ann}(gr M)}$$

Example 19. If $I \triangleleft gr(R)$ is any ideal then for any $x_1, x_2, y_1, y_2 \in I$, $\{x_1 x_2, y_1 y_2\} = x_1 y_1 \{x_2, y_2\} + x_1 \{x_2, y_1\} y_2 + y_1 \{x_1, y_2\} x_2 + \{x_1, y_1\} x_2 y_2 \subset I^2$. So $\{I^2, I^2\} \subset I^2$.

However, if $I = (X_1, Y_1) \triangleleft gr A_1(k)$, then $1 = \{Y_1, X_1\} \notin I = \sqrt{I^2}$

Lemma 9.3. Suppose that A is a commutative Poisson k -algebra and $I \triangleleft A$ is an ideal such that $\{I, I\} \subset I$. Then for every $p : A \rightarrow k$ such that $p(I) = 0$, write $I_p = (I + m^2)/m^2 \subset T_p^* \text{Spec} A$. Then $\{I_p, I_p\}_p = 0$

Suppose that k is a field with $\text{char}(k) = 0$ and M is a $A_1(k)$ -module that is finite dimensional over k (which implies $d(M) = 0$ since the support). Then writing $\theta : A_1(k) \rightarrow \text{End}_k(M)$. Note that $\theta(y)\theta(x) - \theta(x)\theta(y) = \theta(1) = \text{id}_M$. Take trace on both sides and we get $\dim_k(M) = 0$. This says that there are no $A_1(k)$ -modules with $d(M) = 0$. This generalises as follows.

Theorem 9.3. (*Bernstein's Inequality*) Let k be a field of characteristic zero, and let M be a finitely generated, non-zero module over the Weyl algebra $A_n(k)$. Then $d(M) \geq n$

To prove this we need the following lemma from linear algebra:

Lemma 9.4. Let $(-, -)$ be a non-degenerate bilinear form on a finite dimensional k -vector space V , and let W be a subspace of V such that $(W, W) = 0$. Then $\dim W \leq \frac{1}{2} \dim V$

Proof of theorem 9.3. Suppose that M is a nonzero finitely generated $A_n(k)$ -module. By considering $\bar{k} \otimes_k M$ as an $A_n(\bar{k}) = \bar{k} \otimes_k A_n(k)$ -module we may assume that $\bar{k} = k$.

We've seen that if we filter $A_n(k)$ with the filtration by order of differential operators then $A = gr A_n(k) \cong k[X_1, \dots, X_n, Y_1, \dots, Y_n]$ and for every k -point of $\text{Spec} A$, $\{-, -\}_p$ is non-degenerate.

By corollary 9.2, give M a good filtration and let $I = \sqrt{\text{Ann}(gr M)}$. Let $q : A/I \rightarrow k$ be a k -smooth point of $\text{Spec} A/I$ (which gives a k -smooth point p of $\text{Spec} A$ with $p(I) = 0$) so that $d(M) = K \dim(A/I) = \dim_k(T_q^* \text{Spec} A/I) = \dim_k(m_p/(I + m_p^2))$

By Gabber's theorem $\{I, I\} \subset I$, so $\{I_p, I_p\} = 0$ by lemma 9.3. Since $\{-, -\}_p$ is non-degenerate, $\dim_k(I_p) \leq \frac{1}{2} K \dim(A) = \frac{1}{2} (2n) = n$ since $\text{Spec} A$ is smooth everywhere.

Now $I_p = (I + m_p^2)/m_p^2$, so $\dim_k(I_p) = \dim_k(m_p/m_p^2) - \dim_k(m_p/(I + m_p^2)) = 2n - d(M)$. So $d(M) \geq n$ as required. $\square$

The proof of Gabber's theorem is the content of the last lecture, which is non-examinable

9.3 Exercise

Proposition 9.6. *Let k be a field of characteristic p and $n \geq 1$*

1. *Show that the n -th Weyl algebra $A_n(k)$ has a non-zero module that is finite-dimensional over k .*
2. *Describe all the simple modules of $A_n(k)$ in the case that k is algebraically closed.*

Remark 9.7. Note that this can't happen when $\text{char}(k) = 0$ in which case $0 = \text{tr}(y_i x_i - x_i y_i) = \text{tr}(\text{id}_V) = \dim_k(V)$

Proof. (1) Consider $k[t_1, \dots, t_n]/(t_1^p, \dots, t_n^p)$ and $x_i \mapsto \hat{t}_i, y_i \mapsto \partial/\partial t_i$. Can check that $[\partial/\partial t_i, \hat{t}_i] = 1$. Also $y_i(0) = y_i(t_i^p) = p t_i^{p-1} = 0$

(2) Note that in this case $Z(A_n(k)) = k[x_1^p, \dots, y_n^p]$. One can prove that all simple $A_n(k)$ -modules are f.d. By Schur's lemma, $\text{End}_{A_n(k)}(V) = k$. So $Z(A_n(k))$ acts by scalar. In particular, $x_i^p - a_i$ acts on V by 0 for some $a_i \in k$. So the kernel of $Z(A_n(k)) \rightarrow \text{End}_{A_n(k)}(V) \cong k$ contains some $I_{a,b} = (x_i^p - a_i, y_i^p - b_i | i = 1, 2, \dots, n)$ for some $a_i, b_i \in k$. So V is a module of $A_n(k)/I_{a,b}A_n(k)$. We claim it is isomorphic to $M_{p^n}(k)$. First note that $A_n(k)/I_{0,0}A_n(k) \xrightarrow{\sim} A_n(k)/I_{a,b}A_n(k)$ given by $x_i \mapsto x_i - a_i^{\frac{1}{p}}$. Now $A_n(k)/I_{0,0}A_n(k)$ acts on $k[t_1, \dots, t_n]/(t_1^p, \dots, t_n^p)$ as in (1), so $A_n(k)/I_{0,0}A_n(k) \rightarrow \text{End}_{A_n(k)}(k[t_1, \dots, t_n]/(t_1^p, \dots, t_n^p)) \cong M_{p^n}(k)$. One can show this is a bijection. And so it has a unique simple module k^{p^n} . $\square$

Proposition 9.7. *Let $\mathfrak{g} = \mathfrak{sl}_2$ be the $\mathbb{C}$ -Lie algebra with commutator bracket. Let $G = GL_2(\mathbb{C})$ act on $\mathfrak{g}$ by conjugation and on $\mathfrak{g}^*$ by $(g \cdot \theta)(A) = \theta(g^{-1} \cdot A) = \theta(g^{-1}Ag)$ for $\theta \in \mathfrak{g}^*, A \in \mathfrak{g}$*

1. *Show the bilinear form $\mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}; (A, B) \mapsto \text{tr}(AB)$ induces a G -linear bijection $\mathfrak{g} \rightarrow \mathfrak{g}^*$*
2. *Find the G -orbits of $\mathfrak{g}$ and $\mathfrak{g}^*$*
3. *For each $\theta \in \mathfrak{g}^*$ the natural Poisson bracket on $S(\mathfrak{g})$ induces an alternating bilinear form on $\mathfrak{g}$*

$$(x, y)_\theta = \theta([x, y])$$

For a chosen representative θ of each orbit $G \cdot \theta \subset \mathfrak{g}^$ describe $(x, y)_\theta$*

4. *Repeat (1)~(3) for $\mathfrak{g} = \mathfrak{sl}_3$ and $G = GL_3(\mathbb{C})$*

Proof. (1) Define $\gamma : \mathfrak{g} \rightarrow \mathfrak{g}^*$ given by $A \mapsto (B \mapsto \text{tr}(AB))$. It is clearly injective and hence bijective because of dimensions. Now $\gamma(g \cdot A)(B) = \gamma(gAg^{-1})(B) = \text{tr}(gAg^{-1}B) = \text{tr}(BgAg^{-1}) = \text{tr}(Ag^{-1}Bg) = \gamma(A)(g^{-1}Bg) = (g \cdot \gamma(A))(B)$. So it is G -linear.

(2) The G -orbits in $\mathfrak{g}$ are $[\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}], [\begin{pmatrix} \lambda & 0 \\ 0 & -\lambda \end{pmatrix}]$ for each $\lambda \in \mathbb{C}$ and $[\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}]$. From (1), we know the orbits of $\mathfrak{g}^*$ are represented by $\gamma\left(\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\right), \gamma\left(\begin{pmatrix} \lambda & 0 \\ 0 & -\lambda \end{pmatrix}\right)$ and 0. If we write $f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ for the basis of $\mathfrak{g}$, we get $\gamma(e) = f^*, \gamma(f) = e^*$ and $\gamma(\lambda h) = 2\lambda h^*$. So $f^*, 2\lambda h^*$ and 0 represents the G -orbits in $\mathfrak{g}^*$

(3) If $\theta = f^*$, note that $[e, f] = h, [e, h] = -2e$, so $(e, -)_{f^*} = f^*([e, -]) = 0$. Now $[f, h] = 2f$, so $(f, h)_{f^*} = 2$

If $\theta = 2\lambda h^*$, then $(e, f)_\theta = 2\lambda, (e, h)_\theta = 0, (f, h)_\theta = 0$ $\square$

10 Afterthoughts

This is my first (and probably the last) time taking a course on non-commutative rings. The main reason for doing this is that this is a new module in Part iii, and I thought it won't be taught elsewhere. It turns out that Oxford has been offering a similar course for several years. The difference is that this course covers more content, including sections on injective hulls, left ring of fractions, and Goldie's rings and (the general) Goldie's theorem, which I find to be the most difficult part of the entire course because the proofs there are extremely tricky. Although the exam barely touches these topics except for injective modules.

I would say this is a genuinely difficult course because the proofs for most theorems here are very technical. Some of them are particularly not fun because the proofs consist of tricks or smartness and have no interesting analogues like the Kummer pairing in elliptic curves, even though the theorems themselves are interesting and helpful. To prepare for the exam one needs to memorize every trick in the proof of every major theorem, which is extremely painful because there is no similarity or connections between them (just pure techniques!). I don't know how much time one needs to invest to truly master this course, and I will just call it here.